

# TOPICS IN MATHEMATICAL SCIENCE

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### INTRODUCTION TO KOSZUL DUALITY

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## Convention

Throughout the course,  $\mathbb{k}$  is a field. All rings are unital and associative. All modules are right modules unless otherwise specified. When we discuss dimensions, we mean dimensions as  $\mathbb{k}$ -vector spaces.

note for AI: no need to put “Lecture x” everytime, just put lecture title. Restore the numbering on TOC

## 1 Lecture 1: Graded algebras, modules, and duals

### 1.1 Algebras and modules

**Definition 1.1.** A  $\mathbb{k}$ -algebra is a  $\mathbb{k}$ -vector space with an associative unital multiplication that is *bilinear*, meaning linear in each variable.

A right  $A$ -module is understood to be a  $\mathbb{k}$ -vector space  $M$  with a bilinear action satisfying  $(ma)b = m(ab)$  and  $m1_A = m$  for  $m \in M$  and  $a, b \in A$ . A module homomorphism  $f : M \rightarrow N$  is linear and satisfies  $f(ma) = f(m)a$ . A homomorphism of  $\mathbb{k}$ -algebras is a linear map preserving multiplication and the identity.

**Example 1.2.** (1) The matrix ring  $\text{Mat}_n(\mathbb{k})$  is a  $\mathbb{k}$ -algebra. The space of  $1 \times n$  row vectors is a right  $\text{Mat}_n(\mathbb{k})$ -module by matrix multiplication.

(2) Let  $\mathbb{k}A_n$  be the *upper triangular matrix ring*, which is also a  $\mathbb{k}$ -algebra.

$$\mathbb{k}A_n = \begin{pmatrix} \mathbb{k} & \mathbb{k} & \cdots & \mathbb{k} \\ 0 & \mathbb{k} & \cdots & \mathbb{k} \\ 0 & 0 & \ddots & \vdots \\ 0 & \cdots & 0 & \mathbb{k} \end{pmatrix}$$

For those familiar with path algebras,  $\mathbb{k}A_n$  is the path algebra of a *linearly oriented  $A_n$  quiver*, which explains the notation. We use matrix calculations, so no knowledge of quivers is required. Write  $e_{ij}$  for the matrix with 1 in entry  $(i, j)$  and zero elsewhere. Our convention composes paths from left to right:  $e_{ij}$  represents  $i \rightarrow j$ , and  $e_{ij}e_{j\ell} = e_{i\ell}$ .

## 1.2 Categories and functors

We organise modules together with their homomorphisms. A category records both: specifying the objects alone does not specify a category.

**Definition 1.3.** A *category*  $\mathcal{C}$  consists of objects, a set  $\text{Hom}_{\mathcal{C}}(X, Y)$  of *morphisms*  $f : X \rightarrow Y$  for each pair of objects, and composition sending  $f : X \rightarrow Y$  and  $g : Y \rightarrow Z$  to  $g \circ f : X \rightarrow Z$ . Each object  $X$  has an identity  $\text{id}_X : X \rightarrow X$ , and

$$h \circ (g \circ f) = (h \circ g) \circ f, \quad \text{id}_Y \circ f = f = f \circ \text{id}_X \quad \text{for all composable } f, g, h.$$

A morphism  $f : X \rightarrow Y$  is an *isomorphism* if some  $g : Y \rightarrow X$  satisfies  $g \circ f = \text{id}_X$  and  $f \circ g = \text{id}_Y$ ; we then write  $X \cong Y$ . A *full subcategory* of  $\mathcal{C}$  consists of chosen objects of  $\mathcal{C}$ , with all morphisms between them and the same composition.

**Example 1.4.** The category  $\text{Mod } A$  has right  $A$ -modules as objects; its morphism set  $\text{Hom}_A(M, N)$  consists of right  $A$ -linear maps  $M \rightarrow N$ , with the usual composition. Its full subcategory  $\text{mod } A$  consists of finite-dimensional modules. In particular,  $\text{Mod } \mathbb{k}$  and  $\text{mod } \mathbb{k}$  are the categories of all and of finite-dimensional vector spaces, respectively, with linear maps.

**Definition 1.5.** A *functor*  $F : \mathcal{C} \rightarrow \mathcal{D}$  assigns an object  $F(X)$  to each object  $X$  and a morphism  $F(f) : F(X) \rightarrow F(Y)$  to each  $f : X \rightarrow Y$ , such that

$$F(\text{id}_X) = \text{id}_{F(X)}, \quad F(g \circ f) = F(g) \circ F(f) \quad \text{for all composable } f, g.$$

For example, forgetting the  $A$ -action gives a functor  $\text{Mod } A \rightarrow \text{Mod } \mathbb{k}$ : it keeps the underlying vector spaces and linear maps. The *identity functor*  $\text{id}_{\mathcal{C}}$  leaves every object and morphism unchanged; functors compose by applying one after the other. Every functor preserves isomorphisms, since  $F(f^{-1})$  is inverse to  $F(f)$ .

The *opposite category*  $\mathcal{C}^{\text{op}}$  has the same objects as  $\mathcal{C}$ , with  $\text{Hom}_{\mathcal{C}^{\text{op}}}(X, Y) = \text{Hom}_{\mathcal{C}}(Y, X)$  and composition reversed. A *contravariant functor* from  $\mathcal{C}$  to  $\mathcal{D}$  means a functor  $\mathcal{C}^{\text{op}} \rightarrow \mathcal{D}$ . Thus it sends  $f : X \rightarrow Y$  to  $F(f) : F(Y) \rightarrow F(X)$  and satisfies  $F(g \circ f) = F(f) \circ F(g)$ . Duals will provide our main examples.

**Definition 1.6.** For functors  $F, G : \mathcal{C} \rightarrow \mathcal{D}$ , a *natural transformation*  $\eta : F \rightarrow G$  is a family of morphisms  $\eta_X : F(X) \rightarrow G(X)$  such that

$$\begin{array}{ccc} F(X) & \xrightarrow{\eta_X} & G(X) \\ F(f) \downarrow & & \downarrow G(f) \\ F(Y) & \xrightarrow{\eta_Y} & G(Y) \end{array} \quad G(f) \circ \eta_X = \eta_Y \circ F(f) \quad \text{for every } f : X \rightarrow Y. \quad (1.1)$$

If every  $\eta_X$  is an isomorphism,  $\eta$  is a *natural isomorphism*, written  $F \cong G$ .

The square in (1.1) is said to *commute*: its two routes from  $F(X)$  to  $G(Y)$  give the same morphism. For example, for fixed  $c \in \mathbb{k}$ , the maps  $c \text{id}_M : M \rightarrow M$  define a natural transformation  $\text{id}_{\text{Mod } A} \rightarrow \text{id}_{\text{Mod } A}$  because  $f(cm) = cf(m)$  for every module homomorphism  $f : M \rightarrow N$ . It is a natural isomorphism when  $c \neq 0$ .

A functor  $F : \mathcal{C} \rightarrow \mathcal{D}$  is an *equivalence* if there is a functor  $G : \mathcal{D} \rightarrow \mathcal{C}$  with natural isomorphisms  $GF \cong \text{id}_{\mathcal{C}}$  and  $FG \cong \text{id}_{\mathcal{D}}$ . A *duality* between  $\mathcal{C}$  and  $\mathcal{D}$  is an equivalence  $\mathcal{C}^{\text{op}} \rightarrow \mathcal{D}$ . Thus an equivalence recovers objects and maps up to natural isomorphism; a duality also reverses the direction of maps.

add: notation for ungraded module categories, example of  $\mathbb{k}$ -linear dual is a duality functor

### 1.3 Semisimple algebras

**Definition 1.7.** A module  $L$  is **simple** if  $L \neq 0$  and its only submodules are  $0$  and  $L$ . A module is **semisimple** if it is a direct sum of simple modules. An algebra  $B$  is **semisimple** if it is semisimple as a right module  $B_B$  over itself. The **Jacobson radical**  $\text{rad } B$  is the intersection of all maximal right ideals of  $B$ .

We use the following standard facts without proof; see [Bez23, Lecture 2]. Recall that a **division  $\mathbb{k}$ -algebra** is a non-zero  $\mathbb{k}$ -algebra whose non-zero elements are invertible.

**Theorem 1.8 (Artin–Wedderburn).** For a non-zero finite-dimensional  $\mathbb{k}$ -algebra  $B$ ,

$$B \text{ is semisimple} \iff \text{rad } B = 0 \iff B \cong \prod_{i=1}^r \text{Mat}_{n_i}(D_i),$$

where  $r, n_i \geq 1$  and each  $D_i$  is a finite-dimensional division  $\mathbb{k}$ -algebra. If  $\mathbb{k}$  is algebraically closed, then  $D_i = \mathbb{k}$  for every  $i$ . For semisimple  $B$ , every left or right  $B$ -module is semisimple, and every submodule has a complement. In particular, the definition is independent of the chosen side.

**Corollary 1.9.** Every finite-dimensional left or right module over a semisimple algebra  $S$  is a direct summand of  $S^{\oplus n}$  for some  $n$ .

**Proof** A finite generating set gives a surjection  $S^{\oplus n} \twoheadrightarrow M$ . By Theorem 1.8, its kernel has a complement, which maps isomorphically onto  $M$ .  $\square$

**Example 1.10.** The algebras  $\mathbb{k}^{\oplus r}$  and  $\text{Mat}_n(\mathbb{k})$  are semisimple by Theorem 1.8. For  $A = \mathbb{k}A_2$ , the ideal  $J = \mathbb{k}e_{12}$  satisfies  $J^2 = 0$ . If  $L$  is simple, then  $LJ$  is either  $0$  or  $L$ ; the latter would give  $L = LJ^2 = 0$ . Hence every maximal right ideal contains  $J$ . Since  $A/J \cong \mathbb{k} \oplus \mathbb{k}$ , those ideals are  $\mathbb{k}e_{11} \oplus \mathbb{k}e_{12}$  and  $\mathbb{k}e_{12} \oplus \mathbb{k}e_{22}$ . Their intersection is  $J$ , so  $\text{rad } A = \mathbb{k}e_{12} \neq 0$ .

### 1.4 Graded algebras

**Definition 1.11.** Let  $G$  be a group with identity  $e$ . A  **$G$ -graded  $\mathbb{k}$ -algebra** is a  $\mathbb{k}$ -algebra  $A$  with  $1_A \in A_e$  and a decomposition

$$A = \bigoplus_{g \in G} A_g \quad \text{such that} \quad A_g A_h \subset A_{gh} \quad \text{for all } g, h \in G.$$

An element of  $A_g$  is **homogeneous of degree  $g$** .

**Convention.** Unless stated otherwise, a graded algebra is  $\mathbb{Z}$ -graded and concentrated in non-negative degrees.

Note that the decomposition is into a direct sum, so every element of a graded algebra  $A$  can always be written as a *finite* sum of homogeneous elements.

**Example 1.12.** For the upper triangular matrix ring  $\mathbb{k}A_n$  of Example 1.2, the decomposition

$$(\mathbb{k}A_n)_d = \text{span}_{\mathbb{k}}\{e_{ij} \mid j - i = d\}$$

makes  $\mathbb{k}A_n$  a graded algebra.

**Example 1.13.** The algebra  $\mathbb{k}A_3$  also has a  $\mathbb{Z}^{\oplus 2}$ -grading with

$$\deg e_{ii} = (0, 0), \quad \deg e_{12} = (1, 0), \quad \deg e_{23} = (0, 1), \quad \deg e_{13} = (1, 1).$$

The homomorphism  $\mathbb{Z}^{\oplus 2} \rightarrow \mathbb{Z}$ ,  $(p, q) \mapsto p + q$ , gives the grading of Example 1.12. Alternatively, setting  $\deg e_{ij} = 2(j - i)$  gives a  $\mathbb{Z}$ -grading with  $A_1 = 0$  but  $A_2 \neq 0$ .

**Definition 1.14.** A *homomorphism of graded algebras*  $f : A \rightarrow B$  is a  $\mathbb{k}$ -algebra homomorphism with  $f(A_i) \subset B_i$  for all  $i \in \mathbb{Z}$ .

**Definition 1.15.** A graded algebra  $A$  is *locally finite* if  $\dim_{\mathbb{k}} A_i < \infty$  for every  $i \in \mathbb{Z}$ .

## 1.5 Graded modules

**Definition 1.16.** Let  $A = \bigoplus_{g \in G} A_g$  be a  $G$ -graded algebra. A  *$G$ -graded right  $A$ -module* is a right  $A$ -module  $M$  with a direct sum decomposition

$$M = \bigoplus_{g \in G} M_g \quad \text{such that} \quad M_g A_h \subset M_{gh} \quad \text{for all } g, h \in G.$$

An element of  $M_g$  is called *homogeneous of degree  $g$* .

A  $G$ -graded  $A$ -module  $M = \bigoplus_{g \in G} M_g$  is *locally finite* if  $\dim_{\mathbb{k}} M_g < \infty$  for all  $g \in G$ .

Unless specified otherwise, graded modules are  $\mathbb{Z}$ -graded; their degrees need not be non-negative.

The *opposite algebra*  $A^{\text{op}}$  has the same vector space as  $A$ , with multiplication  $a^{\text{op}} b^{\text{op}} = (ba)^{\text{op}}$ . A left  $A$ -module is a right  $A^{\text{op}}$ -module by  $m \cdot a^{\text{op}} = am$ . For a  $\mathbb{Z}$ -graded algebra,  $(A^{\text{op}})_i = A_i$ . Thus  $\text{Mod}(A^{\text{op}})$  is the category of left  $A$ -modules and left  $A$ -linear maps. It is not the opposite category  $(\text{Mod } A)^{\text{op}}$ : the latter keeps right modules as objects and reverses the morphisms.

**Definition 1.17.** An  *$A$ -bimodule* is a vector space  $M$  with  $\mathbb{k}$ -bilinear left and right  $A$ -module actions satisfying  $(am)b = a(mb)$  for all  $a, b \in A$  and  $m \in M$ . It is *graded* if  $M = \bigoplus_{j \in \mathbb{Z}} M_j$  and  $A_i M_j A_k \subset M_{i+j+k}$  for all  $i, j, k \in \mathbb{Z}$ .

The algebra  $A$  is a graded  $A$ -bimodule by multiplication, and each  $A_i$  is an  $A_0$ -bimodule. Bilinearity forces the left and right scalar actions to agree:  $(c1_A)m = cm = m(c1_A)$  for  $c \in \mathbb{k}$ . More precisely, we should have speak of ‘ $\mathbb{k}$ -central  $A$ -bimodule’ instead; but we will always work over a field  $\mathbb{k}$  and so will not make such distinction.

**Example 1.18.** The decomposition  $A = \bigoplus_{i \in \mathbb{Z}} A_i$  makes the right regular module  $A_A$  a graded module, and  ${}_A A_A$  a graded bimodule.

**Definition 1.19.** Let  $M$  be a graded  $A$ -module and let  $d \in \mathbb{Z}$ . The (degree/grading) *shift*  $M(d)$  is the graded  $A$ -module whose underlying  $A$ -module is  $M$ , but whose grading is

$$M(d)_j = M_{j+d}.$$

By Definition 1.19,  $1 \in A_0 = A(-d)_d$ , so  $A(-d)$  has a generator in degree  $d$ . If  $M$  is concentrated in degree 0, then  $M(d)$  is concentrated in degree  $-d$ .

**Example 1.20.** For  $A = \mathbb{k}A_3$  with the grading of Example 1.12, the module  $P_1 := e_{11}A$  consists of matrices supported on the first row. Its basis  $e_{11}, e_{12}, e_{13}$  has degrees 0, 1, 2, respectively; in  $P_1(-2)$ , the same basis has degrees 2, 3, 4.

**Definition 1.21.** Let  $M, N$  be graded  $A$ -modules. A homomorphism  $f : M \rightarrow N$  is *homogeneous of degree  $d$*  if  $f(M_j) \subset N_{j+d}$  for all  $j \in \mathbb{Z}$ . A *homomorphism of graded modules* means a homogeneous  $A$ -module homomorphism of degree 0, also said to *preserve degrees*.

**Example 1.22.** Take  $S = \mathbb{k}e_1 \oplus \mathbb{k}e_2$  in degree 0,  $M = \bigoplus_{i \geq 0} (e_1 S)(-i)$ , and  $N = e_1 S$  in degree 0. The  $S$ -linear map  $M \rightarrow N$  sending the copy of  $e_1$  in every summand to  $e_1$  has a non-zero component in every degree  $-i$ . *write more symbolically, use less words* It is not a finite sum of homogeneous maps.

**Example 1.23.** For  $A = \mathbb{k}A_3$ , left multiplication by  $e_{12}$  yields a map

$$\begin{aligned} e_{22}A &\longrightarrow e_{11}A, \\ \begin{pmatrix} 0 & 0 & 0 \\ 0 & u & v \\ 0 & 0 & 0 \end{pmatrix} &\longmapsto \begin{pmatrix} 0 & u & v \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} \quad (u, v \in \mathbb{k}). \end{aligned}$$

The  $u$  and  $v$  entries have degrees 0, 1 in the source and 1, 2 in the target, and so the map has degree 1.

**Notation.** The space of homomorphisms preserving degrees is

$$\mathrm{Hom}_A^{\mathbb{Z}}(M, N) = \{f \in \mathrm{Hom}_A(M, N) \mid f(M_j) \subset N_j \text{ for all } j \in \mathbb{Z}\}. \quad (1.2)$$

Thus  $\mathrm{Hom}_A^{\mathbb{Z}}(M, N(d))$  consists of homomorphisms  $M \rightarrow N$  of degree  $d$ , since  $N(d)_j = N_{j+d}$ .

We write

$\mathrm{GrMod} A :=$  category of graded right  $A$ -modules with morphism spaces  $\mathrm{Hom}_A^{\mathbb{Z}}(M, N)$ ,  
 $\mathrm{grmod} A :=$  full subcategory of  $\mathrm{GrMod} A$  consisting of locally finite modules.

Forgetting the grading gives a functor  $\mathrm{GrMod} A \rightarrow \mathrm{Mod} A$ . The distinction between its source and target is the grading on objects and the degree condition on morphisms.

For  $d \in \mathbb{Z}$ , the shift of Definition 1.19 is a functor

$$(d) : \mathrm{GrMod} A \rightarrow \mathrm{GrMod} A.$$

It sends  $f : M \rightarrow N$  to the same underlying map  $f(d) : M(d) \rightarrow N(d)$ . Since  $M(d)(-d) = M$  and  $f(d)(-d) = f$ , the shift is an equivalence with inverse  $(-d)$ ; it restricts to an endo-functor on  $\mathrm{grmod} A$ .

**Lemma 1.24.** For graded right  $A$ -modules  $M, N$ , the following inclusion is an equality when  $M$  is finitely generated:

$$\bigoplus_{d \in \mathbb{Z}} \mathrm{Hom}_A^{\mathbb{Z}}(M, N(d)) \subset \mathrm{Hom}_A(M, N).$$

**Proof** For  $f \in \mathrm{Hom}_A(M, N)$ , define  $f_d(m) = f(m)_{j+d}$  when  $m \in M_j$ , and extend linearly. For  $a \in A_i$ , we have  $f_d(ma) = f(ma)_{j+i+d} = f(m)_{j+d}a$ , so  $f_d$  is  $A$ -linear of degree  $d$ . The sum of the homogeneous Hom spaces is direct: on  $M_j$ , distinct degrees have values in distinct summands of  $N$ . If  $M$  is finitely generated, then there is a finite generating set, which can be decomposed to a, still, finite set of *homogeneous* generating elements. Only finitely many  $f_d$  are non-zero on these homogeneous generators, hence on  $M$ , giving  $f = \sum_d f_d$ .  $\square$

## 1.6 Homogeneous ideals and quotients

**Definition 1.25.** Let  $A = \bigoplus_{i \in \mathbb{Z}} A_i$  be a graded algebra. A right ideal  $I \subset A$  is *homogeneous* if

$$I = \bigoplus_{i \in \mathbb{Z}} (I \cap A_i).$$

Equivalently, whenever  $a \in I$  and  $a = a_0 + \cdots + a_N$  is its decomposition into homogeneous components, each  $a_i$  also belongs to  $I$ .

For quotient algebras we need two-sided ideals. If  $I$  is a homogeneous two-sided ideal, then

$$A/I = \bigoplus_{i \in \mathbb{Z}} A_i / (I \cap A_i)$$

is again a graded algebra.

**Example 1.26.** In  $A = \mathbb{k}A_3$  with the grading of Example 1.12, the two-sided ideal  $I = (e_{13}) = \mathbb{k}e_{13}$  is homogeneous. Writing  $\bar{e}_{ij}$  for the image of  $e_{ij}$ ,

$$(A/I)_0 \cong \mathbb{k}^{\oplus 3}, \quad (A/I)_1 = \mathbb{k}\bar{e}_{12} \oplus \mathbb{k}\bar{e}_{23}, \quad (A/I)_d = 0 \quad (d \geq 2).$$

In particular,  $\bar{e}_{12}\bar{e}_{23} = 0$ .

**Example 1.27.** In  $A = \mathbb{k}A_2$ , the subspace

$$I = \mathbb{k}(e_{22} + e_{12}) = \left\{ \begin{pmatrix} 0 & a \\ 0 & a \end{pmatrix} \middle| a \in \mathbb{k} \right\}$$

is a right ideal:  $Ie_{11} = Ie_{12} = 0$  and  $Ie_{22} = I$ . However,  $e_{22}, e_{12} \notin I$ , so  $I$  is not homogeneous. Note that we have  $\bar{e}_{12} = -\bar{e}_{22} \in A/I$  and they lies in  $(A_1 \cap A_0)/I$ , so we do not a direct sum  $A_1/(I \cap A_1) \oplus A_0/(I \cap A_0)$ , i.e. the quotient  $A/I$  does not inherit a grading from  $A$ .

**Definition 1.28.** The **positive degree ideal** of a graded algebra  $A$  is  $A_{>0} := \bigoplus_{i>0} A_i$ .<sup>1</sup>

The subspace  $A_{>0}$  is a homogeneous two-sided ideal. The quotient map

$$\pi : A \longrightarrow A/A_{>0} \cong A_0, \quad \sum_{i \in \mathbb{Z}} a_i \mapsto a_0,$$

makes  $A_0$  a graded  $A$ -bimodule concentrated in degree 0, with  $a \cdot s \cdot b = \pi(a)s\pi(b)$  for  $a, b \in A$  and  $s \in A_0$ .

## 1.7 Tensor products of vector spaces

**Definition 1.29.** For vector spaces  $U, V$ , the **tensor product**  $U \otimes_{\mathbb{k}} V$  is equipped with a bilinear map  $(u, v) \mapsto u \otimes v$  such that every bilinear map  $b : U \times V \rightarrow W$  has a unique linear factorisation:

$$\begin{array}{ccc} U \times V & \xrightarrow{(u,v) \mapsto u \otimes v} & U \otimes_{\mathbb{k}} V \\ & \searrow b & \downarrow \exists! \tilde{b} \\ & & W. \end{array}$$

If both  $U$  and  $V$  has a basis, say,  $(u_i)_{i \in I}$  and  $(v_j)_{j \in J}$  respectively, then we can construct  $U \otimes_{\mathbb{k}} V$  explicitly as the vector space with basis  $\{u_i \otimes v_j \mid i \in I, j \in J\}$ . It is routine to check that any bilinear map  $b : U \times V \rightarrow W$  determines exactly one linear map  $\tilde{b}$  by  $u_i \otimes v_j \mapsto b(u_i, v_j)$ , which gives the desired universal property.

For two spaces satisfying Definition 1.29, the induced maps in both directions are inverse by uniqueness. Thus the construction is independent of the bases up to a unique isomorphism preserving pure tensors.

For finite bases with  $|I| = r$  and  $|J| = s$ , we obtain  $\dim_{\mathbb{k}}(U \otimes_{\mathbb{k}} V) = rs$ . There is a canonical identifications  $(U \otimes_{\mathbb{k}} V) \otimes_{\mathbb{k}} W \cong U \otimes_{\mathbb{k}} (V \otimes_{\mathbb{k}} W)$  and  $\mathbb{k} \otimes_{\mathbb{k}} U \cong U$ . It is also customary to take  $U^{\otimes_{\mathbb{k}} 0} := \mathbb{k}$ .

## 1.8 Graded $\mathbb{k}$ -linear duals

A linear functional on a graded vector space need not have a degree: it may be non-zero on infinitely many homogeneous components. To obtain a graded vector space from taking linear functional, we take only finite sums of homogeneous functionals.

<sup>1</sup>It is also called the augmentation ideal; we avoid that terminology.

**Definition 1.30.** For a vector space  $V$ , its  $\mathbb{k}$ -linear dual is  $D(V) := \text{Hom}_{\mathbb{k}}(V, \mathbb{k})$ . For a graded vector space  $V = \bigoplus_{i \in \mathbb{Z}} V_i$ , its *graded  $\mathbb{k}$ -linear dual* is the graded vector space given by

$$\mathbb{D}(V) = \bigoplus_{i \in \mathbb{Z}} \mathbb{D}(V)_i \quad \text{with} \quad \mathbb{D}(V)_i = D(V_{-i}).$$

For general infinite-dimensional  $V$ , we have  $\mathbb{D}(V) \subset D(V)$ , whereas  $D(V) \cong \prod_{i \in \mathbb{Z}} D(V_i)$ . These spaces coincide when  $V$  is finite-dimensional.

Linear duality reverses the side of a module action. For a right  $A$ -module  $M$  and a left  $A$ -module  $N$ , define

$$(af)(m) := f(ma), \quad (ga)(n) := g(an) \quad (a \in A, f \in D(M), g \in D(N)). \quad (1.3)$$

Here  $m \in M$  and  $n \in N$ ; the first formula makes  $D(M)$  a left module and the second makes  $D(N)$  a right module. For graded modules, the same formulas preserve the graded  $\mathbb{k}$ -linear dual: if  $a \in A_p$  and  $f \in \mathbb{D}(M)_q$ , then  $af$  vanishes outside  $M_{-p-q}$  and hence has degree  $p + q$ .

For a module homomorphism  $u : M \rightarrow N$ , precomposition defines  $D(u) : D(N) \rightarrow D(M)$  by  $D(u)(f) = f \circ u$ . If  $u$  preserves degrees, this restricts to  $\mathbb{D}(u) : \mathbb{D}(N) \rightarrow \mathbb{D}(M)$ , also of degree 0. Thus we obtain functors

$$D : (\text{Mod } A)^{\text{op}} \longrightarrow \text{Mod}(A^{\text{op}}), \quad \mathbb{D} : (\text{GrMod } A)^{\text{op}} \longrightarrow \text{GrMod}(A^{\text{op}}). \quad (1.4)$$

Indeed,  $D(\text{id}_M) = \text{id}_{D(M)}$  and  $D(v \circ u) = D(u) \circ D(v)$  for composable  $u, v$ , and the same identities hold for  $\mathbb{D}$ . Replacing  $A$  by  $A^{\text{op}}$  gives the corresponding functors from left modules to right modules.

**Example 1.31.** Let  $A = \mathbb{k}A_3$  and  $P_1 = e_{11}A$  as in Example 1.20. The functionals  $\varepsilon_{1j}(e_{1\ell}) = \delta_{j\ell}$  for  $1 \leq j, \ell \leq 3$  form a basis of  $\mathbb{D}(P_1)$ , with

$$\deg \varepsilon_{1j} = 1 - j, \quad e_{12}\varepsilon_{12} = \varepsilon_{11}, \quad e_{23}\varepsilon_{13} = \varepsilon_{12}, \quad e_{jj}\varepsilon_{1j} = \varepsilon_{1j}.$$

The actions follow from (1.3).

**do a more proper definition! this is super important!** Recall that a sequence  $0 \rightarrow L \xrightarrow{i} M \xrightarrow{p} N \rightarrow 0$  is *short exact* if  $i$  is injective,  $p$  is surjective, and  $\text{Im } i = \text{Ker } p$ . In  $\text{GrMod } A$  the maps must preserve degrees; exactness can then be checked in each degree.

**Lemma 1.32.** The functor  $\mathbb{D}$  in (1.4) restricts to a duality  $(\text{grmod } A)^{\text{op}} \rightarrow \text{grmod}(A^{\text{op}})$ . Evaluation gives a natural isomorphism

$$\text{ev}_M : M \longrightarrow \mathbb{D}(\mathbb{D}(M)), \quad m \longmapsto (f \mapsto f(m)) \quad (M \in \text{grmod } A). \quad (1.5)$$

Moreover,  $\mathbb{D}$  takes every short exact sequence  $0 \rightarrow L \rightarrow M \rightarrow N \rightarrow 0$  to a short exact sequence  $0 \rightarrow \mathbb{D}(N) \rightarrow \mathbb{D}(M) \rightarrow \mathbb{D}(L) \rightarrow 0$ .

**Proof** The evaluation map (1.5) is an isomorphism in each degree by finite-dimensional linear algebra, and (1.3) makes it  $A$ -linear. For  $u : M \rightarrow N$  of degree 0,  $m \in M$ , and  $f \in \mathbb{D}(N)$ ,

$$(\mathbb{D}(\mathbb{D}(u))(\text{ev}_M(m)))(f) = \text{ev}_M(m)(f \circ u) = f(u(m)) = \text{ev}_N(u(m))(f).$$

Hence  $\mathbb{D}(\mathbb{D}(u)) \circ \text{ev}_M = \text{ev}_N \circ u$ , which is naturality as in (1.1). Evaluation on left modules gives the other natural isomorphism required for a duality. Finally, dualising a short exact sequence in each degree is exact, since every linear functional on a subspace extends to the whole vector space.  $\square$

The same argument without grading gives a duality  $D : (\text{mod } A)^{\text{op}} \rightarrow \text{mod}(A^{\text{op}})$ . Without finite dimensionality for  $D$ , or local finiteness for  $\mathbb{D}$ , evaluation need not be surjective. These duality statements require the finiteness hypotheses.

## 1.9 Left, right, and two-sided duals over $S$

Fix a finite-dimensional semisimple  $\mathbb{k}$ -algebra  $S$ , which is regarded as concentrated in degree 0 when treated as a graded algebra. Besides the  $\mathbb{k}$ -linear dual, we will also work with duals defined by left  $S$ -linear maps, right  $S$ -linear maps, and maps respecting both actions. Their definitions differ, even though the resulting spaces can be identified after a choice of form.

**Definition 1.33.** For a right  $S$ -module  $M$  and a left  $S$ -module  $N$ , define their  $S$ -duals by

$$\begin{aligned} M^* &:= \text{Hom}_S(M, S), & (sf)(m) &:= sf(m), \\ {}^*N &:= \text{Hom}_{S^{\text{op}}}(N, S), & (gs)(n) &:= g(n)s. \end{aligned}$$

Here  $s \in S$ ,  $m \in M$ ,  $n \in N$ ,  $f \in M^*$ , and  $g \in {}^*N$ .

Thus  $M^*$  is a left  $S$ -module and  ${}^*N$  is a right  $S$ -module write this in equation environment  $M \in \text{mod } A \Rightarrow M^* \in \text{mod } A^{\text{op}}$ .

For a right module map  $u : M \rightarrow N$ , define  $u^* : N^* \rightarrow M^*$  by  $u^*(f) = f \circ u$ ; for a left module map  $u : M \rightarrow N$ , the same recipe defines  ${}^*u : {}^*N \rightarrow {}^*M$ . Precomposition gives functors

$$(-)^* : (\text{Mod } S)^{\text{op}} \longrightarrow \text{Mod}(S^{\text{op}}), \quad {}^*(-) : (\text{Mod}(S^{\text{op}}))^{\text{op}} \longrightarrow \text{Mod } S.$$

**Definition 1.34.** should introduce this after tensor product of vector space, and should mention its relation with bimodules immediate after its introduction setup macro for superscript en, like op; always use macro for enveloping algebra notation The enveloping algebra of  $S$  is  $S^{\text{en}} := S \otimes_{\mathbb{k}} S^{\text{op}}$ , with multiplication

$$(s \otimes t^{\text{op}})(s' \otimes (t')^{\text{op}}) = ss' \otimes (t't)^{\text{op}} \quad (s, t, s', t' \in S).$$

after introduce this, there should be some remark about the necessary condition for  $S^{\text{en}}$  to remain semisimple when  $S$  is semisimple; DO NOT CITE ANYTHING and only note as a fact.

after moving definition; use a ‘recall that...’ to begin this part An  $S$ -bimodule is equivalently a right  $S^{\text{en}}$ -module via

$$m \cdot (s \otimes t^{\text{op}}) = tms \quad (s, t \in S, m \in M). \quad (1.6)$$

The first tensor factor acts on the right because we are using right  $S^{\text{en}}$ -modules; the second records the left  $S$ -action. For example,  $(m \cdot (s \otimes t^{\text{op}})) \cdot (s' \otimes (t')^{\text{op}}) = t'tmss' = m \cdot (ss' \otimes (t't)^{\text{op}})$ . A morphism of  $S$ -bimodules is a linear map preserving both actions. Such a map is exactly a right  $S^{\text{en}}$ -module homomorphism under (1.6), so this identifies the category of  $S$ -bimodules with  $\text{Mod } S^{\text{en}}$ . All three spaces  $D(M)$ ,  $M^*$ , and  ${}^*M$  inherit bimodule structures:

$$(s\lambda t)(m) = \lambda(tms), \quad (sft)(m) = sf(tm), \quad (sgt)(m) = g(ms)t, \quad (1.7)$$

where  $\lambda \in D(M)$ ,  $f \in M^*$ ,  $g \in {}^*M$ ,  $s, t \in S$ , and  $m \in M$ .

**Definition 1.35.** The two-sided dual of an  $S$ -bimodule  $M$  is

$$M^{*\text{en}} := \text{Hom}_{S^{\text{en}}}(M, S^{\text{en}}), \quad (sFt)(m) := (s \otimes t^{\text{op}})F(m) \quad (s, t \in S, F \in M^{*\text{en}}, m \in M).$$

In Definition 1.35,  $M$  has the right action (1.6), so  $F(tms) = F(m)(s \otimes t^{\text{op}})$ . The output is a left  $S^{\text{en}}$ -module by multiplication on the values, which gives the displayed bimodule action. The two-sided dual takes values in  $S^{\text{en}}$ , not in  $S$ . On bimodule morphisms it acts by precomposition, as do  $D$ ,  $(-)^*$ , and  ${}^*(-)$ ; each is a contravariant functor on the category of  $S$ -bimodules. It is useful when dualising maps with two module actions at once. The two-sided comparisons are included for reference; the construction of quadratic duals uses only the left and right  $S$ -duals. See [GI20, §2.1] for applications to bimodule resolutions.

## 1.10 Comparing the duals

To identify  $S$ -dual with  $\mathbb{k}$ -dual, we need a suitable linear functional  $S \rightarrow \mathbb{k}$ .

**Definition 1.36.** A *symmetrising form* on a finite-dimensional  $\mathbb{k}$ -algebra  $S$  is a linear map  $\tau : S \rightarrow \mathbb{k}$  satisfying

$$\tau(st) = \tau(ts) \quad \text{for all } s, t \in S, \quad (\tau(st) = 0 \text{ for all } t \in S) \implies s = 0.$$

An algebra admitting such a form is *symmetric*.

Every finite-dimensional semisimple  $\mathbb{k}$ -algebra is symmetric, over an arbitrary field; see [DNN17, Introduction] for this theorem and its original reference. We use the theorem without proof and do not restrict  $S$  to products of matrix algebras over  $\mathbb{k}$ . The forms needed in our basic examples are explicit. we only use the following, so we do not need to cite anything heavy; as instructed, you should NOT cite anything in the beginning lectures. all material should be self-contained. Just say specifically that we will only use the following.

**Example 1.37.** For  $S = \mathbb{k}^{\oplus r}$ , take  $\tau(a_1, \dots, a_r) = \sum_i a_i$ . For  $S = \text{Mat}_n(\mathbb{k})$ , take  $\tau(a) = \text{Tr}(a)$ . Indeed,  $\text{Tr}(ab) = \text{Tr}(ba)$  and  $\text{Tr}(ae_{ji}) = a_{ij}$ , so the pairing is non-degenerate in every characteristic. More generally, the sum of the matrix traces works on every product  $\prod_{j=1}^r \text{Mat}_{n_j}(\mathbb{k})$  in Theorem 1.8. *rewrite more clearly using equation environment for easy reading*

**Proposition 1.38 (Dual comparison, [GI20, §2.1]).** *do not cite anything!* Choose a symmetrising form  $\tau$  on  $S$ . For every finite-dimensional  $S$ -bimodule  $M$ , there are isomorphisms of  $S$ -bimodules

$$\alpha_M : M^* \longrightarrow D(M), \quad f \longmapsto \tau \circ f, \quad \beta_M : {}^*M \longrightarrow D(M), \quad g \longmapsto \tau \circ g. \quad (1.8)$$

There is also an isomorphism of  $S$ -bimodules  $\gamma_M : M^{\text{en}} \rightarrow M^*$  given by

$$\gamma_M(F)(m) = \sum_j a_j \tau(b_j) \quad \text{where} \quad F(m) = \sum_j a_j \otimes b_j^{\text{op}}. \quad (1.9)$$

For fixed  $\tau$ , these are natural isomorphisms of contravariant functors on finite-dimensional  $S$ -bimodules. For example, every bimodule map  $u : M \rightarrow N$  gives a commuting square

$$\begin{array}{ccc} N^* & \xrightarrow{\alpha_N} & D(N) \\ u^* \downarrow & & \downarrow D(u) \\ M^* & \xrightarrow{\alpha_M} & D(M). \end{array} \quad (1.10)$$

The corresponding squares for  $\beta$  and  $\gamma$  also commute. The maps  $\alpha_M$  and  $\beta_M$  also apply to right and left  $S$ -modules, respectively, with the opposite side on the duals.

**Proof** Non-degeneracy makes  $S \rightarrow D(S)$ ,  $a \mapsto (s \mapsto \tau(as))$ , injective, hence bijective because both spaces have the same finite dimension. Thus for  $\lambda \in D(M)$  there exist unique  $f(m), g(m) \in S$  satisfying

$$\tau(f(m)s) = \lambda(ms), \quad \tau(sg(m)) = \lambda(sm) \quad (s \in S, m \in M). \quad (1.11)$$

The inverse of  $S \rightarrow D(S)$  is linear, so  $m \mapsto f(m)$  and  $m \mapsto g(m)$  are linear. Equation (1.11) gives  $f(ms) = f(m)s$ ,  $g(sm) = sg(m)$ , and  $\alpha_M(f) = \beta_M(g) = \lambda$ . Hence  $\alpha_M, \beta_M$  are bijective. For bimodule compatibility, use (1.7) and

$$\tau(sf(tm)) = \tau(f(tm)s) = \tau(f(tms)), \quad \tau(g(ms)t) = \tau(tg(ms)) = \tau(g(tms)).$$

The form  $\tau_{\text{en}}(a \otimes b^{\text{op}}) = \tau(a)\tau(b)$  is symmetrising on  $S^{\text{en}}$ . Indeed,  $\tau_{\text{en}}((a \otimes b^{\text{op}})(c \otimes d^{\text{op}})) = \tau(ac)\tau(db) = \tau(ca)\tau(bd)$ , and its pairing is the tensor product of two non-degenerate pairings. The formula (1.9) satisfies

$$\gamma_M(F)(ms) = \gamma_M(F)(m)s, \quad \gamma_M(sFt) = s\gamma_M(F)t, \quad \alpha_M(\gamma_M(F)) = \tau_{\text{en}} \circ F.$$

The argument (1.11), applied to  $S^{\text{en}}$  and  $\tau_{\text{en}}$ , makes  $F \mapsto \tau_{\text{en}} \circ F$  an isomorphism. Thus  $\gamma_M$  is an isomorphism. For (1.10), both routes send  $f \in N^*$  to  $\tau \circ f \circ u$ ; the checks for  $\beta$  and  $\gamma$  are identical.  $\square$

*Remark 1.39.* The identifications (1.8) and (1.9) depend on  $\tau$ . For instance,  $\tau(a_1, \dots, a_r) = \sum_i c_i a_i$  is symmetrising on  $\mathbb{k}^{\oplus r}$  for any  $c_i \neq 0$ . No semisimplicity assumption on  $S^{\text{en}}$  was used.

**Definition 1.40.** For a graded  $S$ -bimodule  $M$ , define its *graded  $S$ -duals* and *graded two-sided dual* by *setup macro called "grstar" for  $(*)$ , always use macro instead of typing in everytime*

$$M^{(*)} := \bigoplus_{i \in \mathbb{Z}} (M_{-i})^*, \quad ({}^*)M := \bigoplus_{i \in \mathbb{Z}} {}^*(M_{-i}), \quad M^{(*\text{en})} := \bigoplus_{i \in \mathbb{Z}} (M_{-i})^{*\text{en}},$$

where each indicated summand has degree  $i$ . The first construction also applies to graded right  $S$ -modules, and the second to graded left  $S$ -modules.

On degree 0 maps, the graded duals act by precomposition in each degree, so they are contravariant functors on the corresponding categories of graded modules. *write out as equation!!! clarify whether they restricts to lf (locally finite) mods*

Parentheses on the stars distinguish the graded constructions from the full  $S$ -duals. In the notation (1.2),

$$M^{(*)} = \bigoplus_{i \in \mathbb{Z}} \text{Hom}_S^{\mathbb{Z}}(M, S(i)), \quad ({}^*)M = \bigoplus_{i \in \mathbb{Z}} \text{Hom}_{S^{\text{op}}}^{\mathbb{Z}}(M, S(i)).$$

**Corollary 1.41.** For a fixed symmetrising form, the graded dual functors on the category of locally finite graded  $S$ -bimodules are naturally isomorphic: *if it is nat. isom. as functors; then write as such, not applied them on objects*

$$M^{(*)} \cong \mathbb{D}(M) \cong ({}^*)M \cong M^{(*\text{en})}.$$

In particular,  $\mathbb{D}(M) \cong M^{(*)}$  for a graded right  $S$ -module, and  $\mathbb{D}(M) \cong ({}^*)M$  for a graded left  $S$ -module.

**Proof** Apply Proposition 1.38 to  $M_{-i}$  and take the direct sum over  $i \in \mathbb{Z}$ . Every comparison preserves degree because  $\tau$  has degree 0. Naturality is the degreewise version of (1.10).  $\square$

For a graded algebra  $A$  with  $S = A_0$ , Corollary 1.41 transports the  $A$ -action (1.3) to the graded  $S$ -duals of a locally finite graded  $A$ -module:

$$af = \alpha_M^{-1}(a\alpha_M(f)) \quad \text{for right } M, \quad ga = \beta_M^{-1}(\beta_M(g)a) \quad \text{for left } M.$$

Here  $\alpha_M(f) = \tau \circ f$  and  $\beta_M(g) = \tau \circ g$  are the restrictions of (1.8);  $a \in A$ ,  $f \in M^{(*)}$ , and  $g \in ({}^*)M$ . The restrictions to  $S$  agree with Definition 1.33. These remain duals over  $S$ , not duals taking values in  $A$ . The graded  $S$ -bimodules are defined without  $\tau$ , but these transported  $A$ -actions depend on  $\tau$ . The canonical graded  $A$ -module is  $\mathbb{D}(M)$ .

**Example 1.42.** Let  $A = \mathbb{k}A_2$ ,  $S = \mathbb{k}e_{11} \oplus \mathbb{k}e_{22}$ , and  $M = e_{11}A$ . Take  $\tau(e_{11}) = c_1$  and  $\tau(e_{22}) = c_2$  with  $c_1, c_2 \neq 0$ . The right  $S$ -linear maps  $\varphi_0, \varphi_1 \in M^{(*)}$  are specified by  $\varphi_0(e_{11}) = e_{11}$  and  $\varphi_1(e_{12}) = e_{22}$ , with their other basis values zero. Their degrees are 0,  $-1$ , and

$$\alpha_M(e_{12}\varphi_1)(e_{11}) = \alpha_M(\varphi_1)(e_{12}) = c_2, \quad e_{12}\varphi_1 = \frac{c_2}{c_1}\varphi_0.$$

Thus changing the form can change the action on the same vector space. Two choices give isomorphic  $A$ -modules by composing their identifications with  $\mathbb{D}(M)$ .

**Example 1.43.** Let  $S = \mathbb{k}^{\oplus r}$  with coordinate idempotents  $e_1, \dots, e_r$  and the form  $\tau(e_i) = 1$ . When  $S$  is realised as diagonal matrices,  $e_i$  means the matrix unit  $e_{ii}$ . For an  $S$ -bimodule  $M$ , the decompositions

and the reversal of sides are *this is difficult to read, write  $M$  more naturally; see what I did in Example 1.27 and 1.23*

$$M = \bigoplus_{i,j} e_i M e_j, \quad e_j D(M) e_i \cong D(e_i M e_j) \cong e_j M^* e_i \cong e_j (*M) e_i.$$

For the degree 1 piece  $V = \mathbb{k}e_{12}$  of  $\mathbb{k}A_2$ , we have  $e_1 V e_2 = V$ . Its graded  $\mathbb{k}$ -linear dual is spanned by  $\varepsilon$  of degree  $-1$ , with  $\varepsilon(e_{12}) = 1$  and  $e_2 \varepsilon e_1 = \varepsilon$ . The corresponding elements of  $V^*$  and  $*V$  send  $e_{12}$  to  $e_2$  and  $e_1$ , respectively. Thus equal scalar-valued functionals can correspond to different  $S$ -valued maps.

**Lemma 1.44.** *this lemma is difficult to read; too many words clutter together; use equation environment; use proper formula; categorical symbols, etc.* For finite-dimensional right  $S$ -modules  $M$  and left  $S$ -modules  $N$ , evaluation gives natural isomorphisms  $M \cong *(M^*)$  and  $N \cong (*N)^*$ , sending  $m$  to  $(f \mapsto f(m))$  and  $n$  to  $(g \mapsto g(n))$ , respectively. For bimodules, both are bimodule isomorphisms. Consequently,  $(-)^*$  and  $*(-)$  give inverse dualities between  $\text{mod } S$  and  $\text{mod}(S^{\text{op}})$ , without choosing a symmetrising form.

**Proof** By Corollary 1.9, every finite-dimensional  $S$ -module is a direct summand of some  $S^{\oplus n}$ . Evaluation is an isomorphism for  $S$ , hence for such summands; (1.7) gives bimodule compatibility. For  $u : M \rightarrow N$  and  $f \in N^*$ , evaluation at  $m \in M$  gives  $(u^* f)(m) = f(u(m))$ , proving naturality. The left module calculation is the same.  $\square$

## 1.11 Exercises

**Exercise 1.45 (Categories and shifts).** Let  $A = \mathbb{k}A_2$  with the grading of Example 1.12. Show that  $e_{11}A$  and  $(e_{11}A)(1)$  are isomorphic in  $\text{Mod } A$  but not in  $\text{GrMod } A$ . Hint. Compare their degree 1 components. Explain why this does not contradict the fact that the shift functor is an equivalence.

**Exercise 1.46 (Matrix calculations).** Verify the grading on  $\mathbb{k}A_n$  in Example 1.12 using  $e_{ij}e_{pq} = \delta_{jp}e_{iq}$ . Compute  $\text{rad}(\mathbb{k}A_3)$  from the definition. Hint. The strictly upper triangular ideal  $J$  satisfies  $J^3 = 0$ . Apply the argument of Example 1.10 to simple modules and to  $A/J \cong \mathbb{k}^{\oplus 3}$ .

**Exercise 1.47 (Shifts and duals).** For  $A = \mathbb{k}A_3$ , compute the degrees and all non-zero matrix-unit actions on  $\mathbb{D}(e_{22}A)$ . Show that it is not isomorphic to any left module  $Ae_{jj}$ , even after a shift. Hint. Compare dimensions and the actions of  $e_{11}, e_{22}, e_{33}$ .

**Exercise 1.48 (Homogeneous maps).** For the map  $f$  in Example 1.22, write every non-zero homogeneous component  $f_d$  explicitly and prove that no finite sum equals  $f$ .

**Exercise 1.49 (Dependence on the form).** Repeat Example 1.43 with  $\tau(e_i) = c_i \neq 0$ . For  $V = \mathbb{k}e_{12}$ , find the right and left  $S$ -linear maps corresponding to the functional  $\varepsilon(e_{12}) = 1$  under (1.8). Check. Their values on  $e_{12}$  are  $c_2^{-1}e_2$  and  $c_1^{-1}e_1$ , respectively.

## 2 Lecture 2: Tensor and quadratic algebras

### 2.1 Tensor products over an algebra

Let  $A$  be a graded algebra and set  $S = A_0$ . The degree zero algebra  $S$  acts on both sides of  $A_1$ . We therefore need the tensor product over an algebra, not only the tensor product over  $\mathbb{k}$ .

**Definition 2.1.** Let  $M$  be a right  $S$ -module and let  $N$  be a left  $S$ -module. The **tensor product**  $M \otimes_S N$  is the quotient of  $M \otimes_{\mathbb{k}} N$  by the subspace generated by

$$ms \otimes n - m \otimes sn \quad (m \in M, n \in N, s \in S).$$

**Lemma 2.2 (Basic properties of tensor products).** (1) For a vector space  $W$ , a bilinear map  $b : M \times N \rightarrow W$  factors uniquely through  $M \otimes_S N$  if and only if  $b(ms, n) = b(m, sn)$  for all  $m \in M$ ,  $s \in S$ , and  $n \in N$ .

(2) For  $S$ -bimodules  $M, N$ , the tensor product is an  $S$ -bimodule with  $s(m \otimes n)t = sm \otimes nt$ .

(3) For  $S$ -bimodules  $U, V, W$ , there are bimodule isomorphisms

$$\begin{aligned} (U \otimes_S V) \otimes_S W &\xrightarrow{\sim} U \otimes_S (V \otimes_S W), & (u \otimes v) \otimes w &\mapsto u \otimes (v \otimes w), \\ S \otimes_S V &\xrightarrow{\sim} V, & s \otimes v &\mapsto sv, \\ V \otimes_S S &\xrightarrow{\sim} V, & v \otimes s &\mapsto vs. \end{aligned}$$

(4) Tensor products distribute over direct sums in either variable; for example,  $(\bigoplus_i M_i) \otimes_S N \cong \bigoplus_i (M_i \otimes_S N)$ .

**Proof** In (1), Definition 1.29 gives a unique map  $M \otimes_{\mathbb{k}} N \rightarrow W$ ; it descends to the quotient exactly when it kills the relations in Definition 2.1. The action in (2) preserves those relations because the two actions on each bimodule commute. For (3), apply (1) to the displayed maps and their inverses; the unit inverses are  $v \mapsto 1 \otimes v$  and  $v \mapsto v \otimes 1$ . For (4), apply (1) to  $((m_i)_i, n) \mapsto (m_i \otimes n)_i$ ; the inverse is induced by the inclusions  $M_i \rightarrow \bigoplus_i M_i$ . Both composites fix pure tensors. The argument for the other variable is identical.  $\square$

The equality  $ms \otimes_S n = m \otimes_S sn$  allows a scalar from  $S$  to pass between adjacent factors. For an  $S$ -bimodule  $V$ , define

$$V^{\otimes_S 0} := S, \quad V^{\otimes_S i} := \underbrace{V \otimes_S \cdots \otimes_S V}_{i \text{ factors}} \quad (i \geq 1).$$

Lemma 2.2 makes these  $S$ -bimodules independent of parenthesisation and supplies the identifications used in concatenation.

Tensor duality reverses the order of the factors.

**Proposition 2.3.** Let  $S$  be finite-dimensional semisimple and let  $V, W$  be finite-dimensional  $S$ -bimodules. There are bimodule isomorphisms compatible with bimodule maps in  $V, W$ ,

$$\begin{aligned} \theta_D : D(W) \otimes_S V^* &\xrightarrow{\sim} D(V \otimes_S W), \\ \theta_r : W^* \otimes_S V^* &\xrightarrow{\sim} (V \otimes_S W)^*, \\ \theta_\ell : {}^*W \otimes_S {}^*V &\xrightarrow{\sim} {}^*(V \otimes_S W), \\ \theta_{\text{en}} : W^* \otimes_S V^{*\text{en}} &\xrightarrow{\sim} (V \otimes_S W)^{*\text{en}}, \end{aligned} \tag{2.1}$$

defined by

$$\begin{aligned}
\theta_D(\lambda \otimes f)(v \otimes w) &= \lambda(f(v)w), \\
\theta_r(g \otimes f)(v \otimes w) &= g(f(v)w), \\
\theta_\ell(g \otimes f)(v \otimes w) &= f(vg(w)), \\
\theta_{\text{en}}(g \otimes F)(v \otimes w) &= \sum_j g(a_j w) \otimes b_j^{\text{op}}, \quad \text{where } F(v) = \sum_j a_j \otimes b_j^{\text{op}}.
\end{aligned} \tag{2.2}$$

**Proof** The formulas in (2.2) respect the tensor relations and bimodule actions. The maps  $\theta_D$  and  $\theta_r$  are isomorphisms for  $V = S$  as a right module, hence for finite direct sums and direct summands of  $S$ . Corollary 1.9 applies to  $V$ . For  $\theta_\ell$ , apply the same argument to the left module  $W$ . Choose a symmetrising form  $\tau$  as in Definition 1.36, and use  $\gamma_V, \gamma_{V \otimes_S W}$  from (1.9). Then

$$\gamma_{V \otimes_S W} \circ \theta_{\text{en}} = \theta_r \circ (\text{id}_{W^*} \otimes \gamma_V),$$

since both sides evaluate  $g \otimes F$  at  $v \otimes w$  as  $\sum_j g(a_j w) \tau(b_j)$ . Proposition 1.38 makes both  $\gamma$  maps isomorphisms, so  $\theta_{\text{en}}$  is an isomorphism.  $\square$

The maps  $\theta_r, \theta_\ell$  are the evaluations needed for quadratic duality [BGS96, §2.7]. The maps  $\theta_D, \theta_{\text{en}}$  record their compatibility with linear and two-sided duals [GI20, §2.1]; their extra formulas may be kept as reference material.

For a fixed symmetrising form  $\tau$ , let  $\alpha_M(f) = \tau \circ f$  as in (1.8). Then tensor evaluation satisfies

$$\alpha_{V \otimes_S W} \circ \theta_r = \theta_D \circ (\alpha_W \otimes \text{id}_{V^*}).$$

Both sides send  $g \otimes f$  to  $v \otimes w \mapsto \tau(g(f(v)w))$ .

**Example 2.4.** Let  $S = \mathbb{k}^{\oplus r}$  with coordinate idempotents  $e_1, \dots, e_r$ , and suppose  $V = e_i V e_j$  and  $W = e_j W e_\ell$ . By (1.7),

$$V \otimes_S W = e_i(V \otimes_S W)e_\ell, \quad W^* \otimes_S V^* = e_\ell(W^* \otimes_S V^*)e_i.$$

Thus  $\theta_r$  in (2.1) reverses the two outer idempotents.

## 2.2 Generated algebras

Let  $A$  be graded with  $S = A_0$ . Multiplication makes  $A_1$  an  $S$ -bimodule and gives

$$\mu_i : A_1^{\otimes_S i} \longrightarrow A_i, \quad a_1 \otimes \dots \otimes a_i \longmapsto a_1 \dots a_i \quad (i \geq 1), \quad \mu_0 = \text{id}_S. \tag{2.3}$$

**Definition 2.5.** The algebra  $A$  is *generated in degree 1 over  $S$*  if every  $\mu_i$  in (2.3) is surjective.

Equivalently,  $S$  and  $A_1$  generate  $A$  as an algebra.

**Example 2.6.** The algebra  $\mathbb{k}A_n$  of Example 1.12 is generated in degree 1 over  $S = \bigoplus_i \mathbb{k}e_{ii}$ :

$$e_{ij} = e_{i,i+1}e_{i+1,i+2} \dots e_{j-1,j} \quad (1 \leq i < j \leq n).$$

For the grading  $\deg e_{ij} = 2(j-i)$ , the same algebra has  $A_1 = 0$  and  $A_2 \neq 0$ , so it is not generated in degree 1.

### 2.3 Tensor algebras and the universal property

**Definition 2.7.** Let  $S$  be a  $\mathbb{k}$ -algebra and  $V$  an  $S$ -bimodule. The *tensor algebra of  $V$  over  $S$*  is

$$T_S(V) := \bigoplus_{i \in \mathbb{Z}} T_S(V)_i, \quad \text{where} \quad T_S(V)_i = \begin{cases} V^{\otimes i}, & i \geq 1, \\ S, & i = 0, \\ 0, & i < 0, \end{cases}$$

with multiplication given by concatenation of tensors and the given  $S$ -actions.

The left and right  $S$ -actions determine which factors can be concatenated. Juxtaposition  $v_1 \cdots v_d$  in  $T_S(V)$  denotes the tensor  $v_1 \otimes_S \cdots \otimes_S v_d$ ; in a specified quotient it denotes the image of that tensor.

**Lemma 2.8.** Let  $S = \bigoplus_{i=1}^r \mathbb{k}e_i$  with coordinate idempotents and let  $V, W$  be  $S$ -bimodules. There is a bimodule isomorphism

$$V \otimes_S W \cong \bigoplus_{i,j,\ell} e_i V e_j \otimes_{\mathbb{k}} e_j W e_\ell. \quad (2.4)$$

Consequently, after choosing a basis of each  $e_i V e_j$ , a basis of  $V^{\otimes d}$  for  $d \geq 1$  consists of the words of length  $d$  whose adjacent idempotents match.

**Proof** The mutually orthogonal idempotents sum to 1, so

$$V = \bigoplus_{i,j} e_i V e_j, \quad (e_i V e_j) \otimes_S (e_p W e_\ell) = 0 \quad \text{if } j \neq p. \quad (2.5)$$

For the second identity,  $v \otimes w = v e_j \otimes w = v \otimes e_j w = 0$ . The inclusions of the matching components define the map from right to left in (2.4). Its inverse is

$$v \otimes_S w \longmapsto \sum_{i,j,\ell} e_i v e_j \otimes_{\mathbb{k}} e_j w e_\ell.$$

For  $s = \sum_j s_j e_j \in S$ , replacing  $(v, w)$  by  $(vs, w)$  or  $(v, sw)$  multiplies the  $j$ th summand by the same scalar  $s_j$ , so the inverse is well-defined. Both composites are the identity by (2.5). Iterating (2.4) and using the vector-space tensor basis proves the assertion about words.  $\square$

**Example 2.9.** Let  $S = \mathbb{k}^{\oplus 3}$  with coordinate idempotents  $e_1, e_2, e_3$ , and let  $V = \mathbb{k}a \oplus \mathbb{k}b$  have actions  $e_1 a e_2 = a$  and  $e_2 b e_3 = b$ , with all other idempotent actions zero. By Lemma 2.8,

$$V \otimes_S V = \mathbb{k}(a \otimes b), \quad V^{\otimes 3} = 0, \quad T_S(V) = S \oplus \mathbb{k}a \oplus \mathbb{k}b \oplus \mathbb{k}ab.$$

Here  $ab = a \otimes b$ . The assignments  $e_i \mapsto e_{ii}$ ,  $a \mapsto e_{12}$ , and  $b \mapsto e_{23}$  give  $T_S(V) \cong \mathbb{k}A_3$ ; the tensor  $ab$  maps to  $e_{13}$ .

**Proposition 2.10 (Universal property of the tensor algebra).** Let  $\iota_S : S \rightarrow T_S(V)$  and  $\iota_V : V \rightarrow T_S(V)$  be the canonical inclusions. Given a unital  $\mathbb{k}$ -algebra homomorphism  $\varphi : S \rightarrow B$  and an  $S$ -bimodule map  $f : V \rightarrow B$ , where  $sbt = \varphi(s)b\varphi(t)$  on  $B$ , there is a unique  $\mathbb{k}$ -algebra homomorphism  $\tilde{f} : T_S(V) \rightarrow B$  making the diagram commute:

$$\begin{array}{ccccc} V & \xrightarrow{\iota_V} & T_S(V) & \xleftarrow{\iota_S} & S \\ & \searrow f & \downarrow \exists! \tilde{f} & \swarrow \varphi & \\ & & B & & \end{array} \quad (2.6)$$

**Proof** Define  $\tilde{f}$  linearly by

$$\tilde{f}(s) = \varphi(s), \quad \tilde{f}(v_1 \otimes_S \cdots \otimes_S v_i) = f(v_1) \cdots f(v_i) \quad (s \in S, i \geq 1). \quad (2.7)$$

The identity  $f(svt) = \varphi(s)f(v)\varphi(t)$  makes  $\tilde{f}$  well-defined on tensor products and compatible with multiplication. Every homomorphism satisfying (2.6) has the values in (2.7), proving uniqueness.  $\square$

**Corollary 2.11.** *If  $A$  is generated in degree 1 over  $S = A_0$ , there is a surjective graded algebra homomorphism  $\mu : T_S(A_1) \twoheadrightarrow A$  whose degree  $i$  component is  $\mu_i$  from (2.3).*

**Proof** Apply Proposition 2.10 to the inclusions  $S \rightarrow A$  and  $A_1 \rightarrow A$ . Surjectivity of each  $\mu_i$  gives surjectivity of  $\mu$ .  $\square$

**Definition 2.12.** *For the map  $\mu$  in Corollary 2.11, the homogeneous two-sided ideal  $I := \text{Ker } \mu$  is the ideal of relations.*

Writing  $V = A_1$ , we obtain  $A \cong T_S(V)/I$  with  $I_0 = I_1 = 0$ . The bimodule  $V$  specifies the generators and their left and right  $S$ -actions;  $I$  specifies the relations between their products.

## 2.4 Quadratic algebras

**Definition 2.13** ([BGS96, Definition 1.2.2]). *Let  $S$  be finite-dimensional semisimple,  $V$  a finite-dimensional  $S$ -bimodule, and  $R \subset V \otimes_S V$  an  $S$ -sub-bimodule. A graded algebra  $A$  is **quadratic** if it is isomorphic, as a graded algebra, to  $T_S(V)/(R)$  for such  $S, V, R$ , where  $(R)$  denotes the two-sided ideal generated by  $R$  and  $V$  has degree 1.*

**Proposition 2.14.** *A quadratic algebra  $A = T_S(V)/(R)$  has  $A_0 = S$ ,  $A_1 = V$ , and  $A_2 = (V \otimes_S V)/R$ , is locally finite, and is generated in degree 1 over  $S$ .*

**Proof** The ideal  $(R)$  has no components in degrees 0 and 1. Its degree 2 component is exactly  $R$ . Each  $A_i$  is a quotient of the finite-dimensional space  $V^{\otimes_S i}$ , and multiplication maps  $V^{\otimes_S i}$  onto  $A_i$ .  $\square$

Thus a quadratic presentation recovers  $S \cong A_0$ ,  $V \cong A_1$ , and  $R \cong \text{Ker}(A_1 \otimes_{A_0} A_1 \rightarrow A_2)$ . In particular, the kernel of the canonical map  $T_{A_0}(A_1) \twoheadrightarrow A$  is generated by its degree 2 component. Quadraticity depends on the grading: for  $n \geq 2$ , the grading  $\deg e_{ij} = 2(j - i)$  on  $\mathbb{k}A_n$  is not quadratic, since it has no degree 1 generators.

**Example 2.15.** *Let  $S = \bigoplus_{i=1}^n \mathbb{k}e_{ii}$  and  $V = \bigoplus_{i=1}^{n-1} \mathbb{k}e_{i,i+1}$  inside  $\mathbb{k}A_n$ . By Lemma 2.8, the basis tensors of positive length are*

$$e_{i,i+1} \otimes_S e_{i+1,i+2} \otimes_S \cdots \otimes_S e_{j-1,j} \longmapsto e_{ij} \quad (1 \leq i < j \leq n).$$

*Thus multiplication gives  $T_S(V) \cong \mathbb{k}A_n$ , a quadratic algebra with  $R = 0$ .*

**Example 2.16.** *For  $S, V, a, b$  from Example 2.9, the quotient*

$$T_S(V)/(ab) \cong \mathbb{k}A_3/(e_{13})$$

*is quadratic with  $R = \mathbb{k}(a \otimes b)$ . Its basis is  $e_1, e_2, e_3, a, b$ , in degrees 0, 0, 0, 1, 1, as in Example 1.26.*

**Example 2.17.** *Let  $S = \mathbb{k}^{\oplus 2}$  with coordinate idempotents  $e_1, e_2$ , and let  $V = \mathbb{k}a \oplus \mathbb{k}b$  satisfy  $e_1 a e_2 = a$  and  $e_2 b e_1 = b$ , with all other idempotent actions zero. By Lemma 2.8, the alternating words in  $a, b$  form the basis of positive degree tensors in  $T_S(V)$ . The quadratic algebra  $A = T_S(V)/(ab)$  has*

$$A = S \oplus \mathbb{k}a \oplus \mathbb{k}b \oplus \mathbb{k}ba, \quad \dim_{\mathbb{k}} A_0 = 2, \quad \dim_{\mathbb{k}} A_1 = 2, \quad \dim_{\mathbb{k}} A_2 = 1.$$

*Every alternating word of length at least 3 contains  $ab$ , so all higher components vanish.*

**Example 2.18.** *Let  $B = \mathbb{k}A_4$  and  $A = B/(e_{14})$ , with the grading of Example 1.12. The ideal  $I = (e_{14}) = \mathbb{k}e_{14}$  has  $I_2 = 0$  but  $I_3 \neq 0$ . Since  $e_{14} = e_{12}e_{23}e_{34}$ , the algebra  $A$  is generated in degree 1 with a relation of degree 3 and no relations of degree 2. By Proposition 2.14, any quadratic presentation would have the same generators and zero quadratic relation space, hence would give  $\mathbb{k}A_4$  itself. Thus  $A$  is not quadratic.*

**Example 2.19 (Polynomial and exterior algebras).** For the classical comparison, take  $S = \mathbb{k}$  and  $V = \bigoplus_{i=1}^n \mathbb{k}x_i$ . The *symmetric algebra* and *exterior algebra* are

$$\begin{aligned} \text{Sym}(V) &= T_{\mathbb{k}}(V)/(x_i x_j - x_j x_i \mid i < j) \cong \mathbb{k}[x_1, \dots, x_n], \\ \bigwedge V &= T_{\mathbb{k}}(V)/(v^2 \mid v \in V). \end{aligned} \tag{2.8}$$

Both are quadratic. The polynomial basis allows repetitions; the exterior basis consists of 1 and  $x_{i_1} \cdots x_{i_d}$  with  $i_1 < \cdots < i_d$ . The exterior relations are equivalently  $x_i^2 = 0$  for  $1 \leq i \leq n$  and  $x_i x_j + x_j x_i = 0$  for  $i < j$ , in every characteristic.

The bases in Example 2.19 can be checked directly. For  $\text{Sym}(V)$ , commuting adjacent generators gives the ordered monomials, which are independent under the map to the polynomial ring. For  $\bigwedge V$ , the relations reduce every word to a multiple of a strictly increasing word or to zero. To prove independence in degree  $d$ , fix  $i_1 < \cdots < i_d$  and send  $v_1 \otimes \cdots \otimes v_d$  to the determinant of their coordinates in rows  $i_1, \dots, i_d$ . This functional vanishes on tensors with two equal adjacent factors, hence on the degree  $d$  part of  $(v^2 \mid v \in V)$ . On increasing basis words it is 1 on  $x_{i_1} \cdots x_{i_d}$  and 0 on the others, proving independence in every characteristic.

## 2.5 Hilbert series

A Hilbert series records the dimension of each homogeneous component.

**Definition 2.20.** Let  $M = \bigoplus_{j \in \mathbb{Z}} M_j$  be a graded vector space with  $\dim_{\mathbb{k}} M_j < \infty$  for all  $j$  and  $M_j = 0$  for all  $j \ll 0$ . Its *Hilbert series* is the formal Laurent series  $h_M(t) := \sum_{j \in \mathbb{Z}} (\dim_{\mathbb{k}} M_j) t^j$ .

For a locally finite graded algebra  $A$ ,  $h_A(t)$  refers to its underlying graded vector space; the series does not record multiplication.

**Lemma 2.21.** Let  $L, M, N$  be locally finite graded vector spaces concentrated in degrees bounded below.

- (1) A graded isomorphism  $M \cong N$  implies  $h_M(t) = h_N(t)$ .
- (2) An exact sequence  $0 \rightarrow L \rightarrow M \rightarrow N \rightarrow 0$  with degree 0 maps gives  $h_M(t) = h_L(t) + h_N(t)$ .
- (3) For every  $d \in \mathbb{Z}$ ,  $h_{M(d)}(t) = t^{-d} h_M(t)$ .

**Proof** The first two statements follow from dimensions in each degree. For shifts,

$$h_{M(d)}(t) = \sum_j \dim_{\mathbb{k}}(M_{j+d}) t^j = t^{-d} \sum_i \dim_{\mathbb{k}}(M_i) t^i. \quad \square$$

**Example 2.22.** For the upper triangular algebra in Example 1.12, counting matrix entries on each diagonal gives

$$h_{\mathbb{k}A_n}(t) = n + (n-1)t + \cdots + t^{n-1}, \quad h_{\mathbb{k}A_3/(e_{13})}(t) = 3 + 2t.$$

The second equality also follows by subtracting  $h_{(e_{13})}(t) = t^2$  from  $h_{\mathbb{k}A_3}(t)$ , as in Lemma 2.21(2). For  $A = T_S(V)/(ab)$  in Example 2.17, the basis  $e_1, e_2, a, b, ba$  gives  $h_A(t) = 2 + 2t + t^2$ .

**Example 2.23.** For the algebras in Example 2.19, each polynomial generator may occur any number of times, whereas each exterior generator occurs at most once. Thus

$$h_{\text{Sym}(V)}(t) = \frac{1}{(1-t)^n}, \quad h_{\bigwedge V}(t) = (1+t)^n \quad \text{where } n = \dim_{\mathbb{k}} V.$$

**Definition 2.24.** For graded vector spaces  $V, W$ , the *total grading* on their tensor product is

$$(V \otimes_{\mathbb{k}} W)_n = \bigoplus_{i+j=n} V_i \otimes_{\mathbb{k}} W_j.$$

**Lemma 2.25.** *For locally finite graded vector spaces  $V, W$  concentrated in degrees bounded below,  $h_{V \otimes_{\mathbb{k}} W}(t) = h_V(t)h_W(t)$ .*

**Proof** The coefficient of  $t^n$  on either side is  $\sum_{i+j=n} \dim_{\mathbb{k}} V_i \dim_{\mathbb{k}} W_j$ . Only finitely many summands are non-zero because both gradings are bounded below.  $\square$

## 2.6 The degree 2 relations of a quotient

Let  $S$  be finite-dimensional semisimple and  $V$  a finite-dimensional  $S$ -bimodule. Let  $A = T_S(V)/I$  with  $I_0 = I_1 = 0$  as in Definition 2.12. In degree 2,

$$A_2 \cong (V \otimes_S V)/I_2, \quad \text{where } I_2 = I \cap (V \otimes_S V).$$

**Definition 2.26.** *The [quadratic part](#) of  $A$  is  $qA := T_S(V)/(I_2)$ .*

The inclusion  $(I_2) \subset I$  induces a surjective graded homomorphism  $\pi_A : qA \twoheadrightarrow A$ , and

$$\pi_A \text{ is an isomorphism} \iff I = (I_2) \iff A \text{ is quadratic.}$$

**Example 2.27.** *For  $A = \mathbb{k}A_4/(e_{14})$  from Example 2.18, we have  $I_2 = 0$ , so*

$$qA = \mathbb{k}A_4, \quad \pi_A : \mathbb{k}A_4 \twoheadrightarrow \mathbb{k}A_4/(e_{14}), \quad \text{Ker } \pi_A = \mathbb{k}e_{14}.$$

*Thus  $qA$  does not impose the cubic relation  $e_{12}e_{23}e_{34} = 0$ .*

## 2.7 Orthogonal complements and quadratic duality

**Definition 2.28.** *For an  $S$ -sub-bimodule  $U$  of a finite-dimensional  $S$ -bimodule  $M$ , define the [right orthogonal complement](#) and [left orthogonal complement](#) by*

$$U^\perp := \{f \in M^* \mid f(U) = 0\}, \quad {}^\perp U := \{g \in {}^*M \mid g(U) = 0\}.$$

Both  $U^\perp$  and  ${}^\perp U$  are  $S$ -sub-bimodules by (1.7). Fix a symmetrising form  $\tau$  as in Definition 1.36. For the comparison maps  $\alpha_M(f) = \tau \circ f$  and  $\beta_M(g) = \tau \circ g$  in (1.8),

$$\alpha_M(U^\perp) = \beta_M({}^\perp U) = \{\lambda \in D(M) \mid \lambda(U) = 0\}. \quad (2.9)$$

Indeed, if  $\alpha_M(f)(U) = 0$ , then

$$\tau(f(u)s) = \tau(f(us)) = 0 \quad \text{for all } s \in S, u \in U,$$

so non-degeneracy of  $\tau$  gives  $f(U) = 0$ . For  $\beta_M$ , use  $\tau(sg(u)) = \tau(g(su))$  instead.

**Example 2.29.** *For  $\text{Sym}(V)$  and  $\bigwedge V$  in Example 2.19, take  $V = \mathbb{k}x \oplus \mathbb{k}y$ . The relation spaces are*

$$\begin{aligned} R_{\text{Sym}(V)} &= \mathbb{k}(x \otimes y - y \otimes x), \\ R_{\bigwedge V} &= \mathbb{k}(x \otimes x) \oplus \mathbb{k}(y \otimes y) \oplus \mathbb{k}(x \otimes y + y \otimes x). \end{aligned}$$

*Let  $x^*, y^* \in V^*$  be the coordinate functionals. Identifying  $(V \otimes_{\mathbb{k}} V)^*$  with  $V^* \otimes_{\mathbb{k}} V^*$  using  $\theta_r$  from (2.2),*

$$\begin{aligned} R_{\text{Sym}(V)}^\perp &= \text{span}_{\mathbb{k}}\{x^* \otimes x^*, y^* \otimes y^*, x^* \otimes y^* + y^* \otimes x^*\}, \\ R_{\bigwedge V}^\perp &= \mathbb{k}(x^* \otimes y^* - y^* \otimes x^*). \end{aligned}$$

*Thus taking orthogonal complements exchanges the symmetric and exterior relation spaces.*

## 2.8 Exercises

**Exercise 2.30.** Let  $A = \mathbb{K}A_4/(e_{13}, e_{24})$  with the grading of Example 1.12. Find a homogeneous basis, compute  $h_A(t)$ , and write  $A = T_S(V)/(R)$  with explicit  $S, V, R$ . Hint. Use Lemma 2.8; the ideal contains  $e_{14}$  as well as  $e_{13}, e_{24}$ . The Hilbert series is  $4 + 3t$ .

**Exercise 2.31.** For the bimodule  $V$  in Example 2.17, find a basis of each  $T_S(V)_d$  and show that  $h_{T_S(V)}(t) = 2/(1-t)$ . Compare with the quotient  $T_S(V)/(ab)$ . Hint. Count alternating words separately in each degree, including degree 0.

**Exercise 2.32.** For the algebras in Example 2.19, prove

$$\dim_{\mathbb{K}} \operatorname{Sym}(V)_2 = \frac{n(n+1)}{2}, \quad \dim_{\mathbb{K}} (\bigwedge V)_2 = \frac{n(n-1)}{2} \quad \text{where } n = \dim_{\mathbb{K}} V.$$

Hint. Count the ordered pairs allowed in the two bases of Example 2.19.

### 3 Lecture 3: The quadratic dual

#### 3.1 Generators and orthogonal relations

Throughout this lecture,  $S$  is a finite-dimensional semisimple  $\mathbb{k}$ -algebra and  $A = T_S(V)/(R)$  is quadratic, with  $V$  finite-dimensional and  $R \subset V \otimes_S V$  an  $S$ -sub-bimodule. The quadratic dual records which linear functionals on products of two generators vanish on the relations. Proposition 3.2 makes the exchange explicit: its degree 2 component is the  $S$ -dual of the relation space of  $A$ . Recall that  $V^* = \text{Hom}_S(V, S)$  uses right  $S$ -linear maps, whereas  ${}^*V = \text{Hom}_{S^{\text{op}}}(V, S)$  uses left  $S$ -linear maps. The tensor evaluations in (2.2) are

$$\begin{aligned}\theta_r(g \otimes f)(v \otimes w) &= g(f(v)w), \quad f, g \in V^*, \\ \theta_\ell(g \otimes f)(v \otimes w) &= f(vg(w)), \quad f, g \in {}^*V.\end{aligned}\tag{3.1}$$

Both reverse the order of the factors. The quadratic dual replaces  $V$  by an  $S$ -dual and  $R$  by the maps vanishing on  $R$ .

**Convention.** We identify  $V^* \otimes_S V^*$  with  $(V \otimes_S V)^*$  via  $\theta_r$ , and  ${}^*V \otimes_S {}^*V$  with  ${}^*(V \otimes_S V)$  via  $\theta_\ell$ . Thus  $R^\perp$  and  ${}^\perp R$  in tensor algebras mean the inverse images of the orthogonal complements in Definition 2.28 under these evaluations.

**Definition 3.1** ([BGS96, §2.8]). The *right quadratic dual* and *left quadratic dual* of  $A$  are

$$\begin{aligned}A^\dagger &:= T_S(V^*)/(R^\perp), \quad R^\perp = \{F \in V^* \otimes_S V^* \mid \theta_r(F)(R) = 0\}, \\ {}^\dagger A &:= T_S({}^*V)/({}^\perp R), \quad {}^\perp R = \{F \in {}^*V \otimes_S {}^*V \mid \theta_\ell(F)(R) = 0\}.\end{aligned}\tag{3.2}$$

The generators  $V^*$  and  ${}^*V$  have degree 1.

We use  $A^\dagger$  unless a side is specified. The construction uses the ordinary  $S$ -dual of  $V$ , placed in degree 1, not the graded  $S$ -dual  $V^{(*)}$ , which has degree  $-1$ . It is independent of a chosen presentation:  $V = A_1$  and  $R = \text{Ker}(A_1 \otimes_S A_1 \rightarrow A_2)$  are intrinsic to  $A$ . An isomorphism *over  $S$*  means one restricting to  $\text{id}_S$  on degree 0.

**Proposition 3.2.** The algebra  $A^\dagger$  is quadratic and

$$A_0^\dagger = S, \quad A_1^\dagger = V^*, \quad A_2^\dagger \cong R^*.$$

In particular,  $\dim_{\mathbb{k}} A_2^\dagger = \dim_{\mathbb{k}} R$ .

**Proof** Restriction to  $R$  gives an exact sequence of  $S$ -bimodules

$$0 \longrightarrow R^\perp \longrightarrow V^* \otimes_S V^* \xrightarrow{F \mapsto \theta_r(F)|_R} R^* \longrightarrow 0.\tag{3.3}$$

Surjectivity follows from Theorem 1.8:  $R$  has a complement as a right  $S$ -module, so a right  $S$ -linear map  $R \rightarrow S$  extends by zero on that complement. Now use Proposition 2.14; the dimension statement follows from Proposition 1.38.  $\square$

Adding relations has the opposite effect after dualising. For  $S$ -sub-bimodules  $R \subset R' \subset V \otimes_S V$ ,

$$R \subset R' \implies (R')^\perp \subset R^\perp \implies (T_S(V)/(R'))^\dagger \twoheadrightarrow (T_S(V)/(R))^\dagger.$$

#### 3.2 Recovering the original algebra

**Lemma 3.3.** Let  $U \subset M$  be an  $S$ -sub-bimodule of a finite-dimensional  $S$ -bimodule. Under the evaluation isomorphisms of Lemma 1.44,

$${}^\perp(U^\perp) = U, \quad ({}^\perp U)^\perp = U.$$

**Proof** Every  $u \in U$  is annihilated by every  $f \in U^\perp$ , so  $U \subset {}^\perp(U^\perp)$ . If  $m \notin U$ , then  $m + U \neq 0$  in  $M/U$ . By Corollary 1.9,  $M/U$  embeds as a direct summand of some  $S^{\oplus n}$  as a right module. At least one coordinate of the image of  $m + U$  is non-zero, giving a right  $S$ -linear map  $M/U \rightarrow S$  non-zero on  $m + U$ . Its composite with  $M \rightarrow M/U$  belongs to  $U^\perp$  and does not vanish on  $m$ . Hence  $m \notin {}^\perp(U^\perp)$ . The proof of  $({}^\perp U)^\perp = U$  uses left modules instead.  $\square$

**Theorem 3.4.** *Left and right quadratic duality are canonically inverse:*

$${}^!(A^!) \cong A \cong ({}^!A)^! \quad \text{as graded algebras over } S.$$

**Proof** *Generators.* Write  $\text{ev}_V : V \rightarrow {}^*(V^*)$ ,  $\text{ev}_V(v)(f) = f(v)$ , for the isomorphism in Lemma 1.44. It identifies the generators of  $A$  and  ${}^!(A^!)$ .

*Relations.* We must check that the induced identification on tensors agrees with evaluation on  $V \otimes_S V$ . The square

$$\begin{array}{ccc} V \otimes_S V & \xrightarrow{\text{ev}_V \otimes \text{ev}_V} & {}^*(V^*) \otimes_S {}^*(V^*) \\ \text{ev}_V \otimes_S V \downarrow & & \downarrow \theta_\ell \\ {}^*((V \otimes_S V)^*) & \xrightarrow{F \mapsto F \circ \theta_r} & {}^*(V^* \otimes_S V^*) \end{array} \quad (3.4)$$

commutes, where the right vertical map is (3.1) applied to the bimodule  $V^*$ . Indeed, for  $v, w \in V$  and  $f, g \in V^*$ , the right  $S$ -action on  $V^*$  in (1.7) gives

$$\begin{aligned} \theta_\ell(\text{ev}_V(v) \otimes \text{ev}_V(w))(g \otimes f) &= \text{ev}_V(w)(gf(v)) \\ &= g(f(v)w) = \theta_r(g \otimes f)(v \otimes w). \end{aligned}$$

Thus, using the identifications (3.1), the relations  ${}^\perp(R^\perp)$  of  ${}^!(A^!)$  correspond to the double annihilator of  $R$  in  $V \otimes_S V$ . Lemma 3.3 identifies that annihilator with  $R$ , so (3.2) gives  ${}^!(A^!) \cong A$ . Interchanging left and right gives  $({}^!A)^! \cong A$ .  $\square$

**Proposition 3.5.** *A choice of symmetrising form  $\tau : S \rightarrow \mathbb{k}$  gives isomorphisms  $A^! \cong {}^!A$  and  $(A^!)^! \cong A$  of graded algebras over  $S$ .*

**Proof** The maps  $\alpha_V(f) = \tau \circ f$  and  $\beta_V(g) = \tau \circ g$  in (1.8) give a bimodule isomorphism  $\beta_V^{-1}\alpha_V : V^* \rightarrow {}^*V$ . For  $f, g \in V^*$ , write  $f' = \beta_V^{-1}\alpha_V(f)$  and  $g' = \beta_V^{-1}\alpha_V(g)$ . Then

$$\begin{aligned} \tau(f'(vg'(w))) &= \tau(f(v)g'(w)) = \tau(g'(f(v)w)) \\ &= \tau(g(f(v)w)) \quad (v, w \in V). \end{aligned}$$

By (2.9),  $(\beta_V^{-1}\alpha_V)^{\otimes_S 2}$  therefore identifies  $R^\perp$  with  ${}^\perp R$ . It induces  $A^! \cong {}^!A$  on the presentations (3.2). Apply the same comparison to  $A^!$  and then Theorem 3.4 to obtain  $(A^!)^! \cong ({}^!A)^! \cong A$ .  $\square$

Theorem 3.4 needs no choice of form. Proposition 3.5 permits the usual statement that quadratic duality is an involution, but the identification of left and right duals uses  $\tau$ .

### 3.3 Tensor algebras and monomial relations

**Example 3.6.** *Taking  $R = 0$  or  $R = V \otimes_S V$  in (3.2) gives*

$$\begin{aligned} (T_S(V))^! &= T_S(V^*)/(V^* \otimes_S V^*) \cong S \oplus V^*, \\ (T_S(V)/(V \otimes_S V))^! &= T_S(V^*). \end{aligned}$$

*In  $S \oplus V^*$ ,  $(V^*)^2 = 0$  and  $(s, f)(t, g) = (st, sg + ft)$ . Thus a tensor algebra with no relations is dual to an algebra in which every product of two positive degree elements is zero.*

For the rest of this subsection, take  $S = \bigoplus_{i=1}^r \mathbb{k}e_i$ , with  $e_i e_j = \delta_{ij} e_i$  and  $\sum_i e_i = 1$ . Choose a basis  $\mathcal{B}_{ij}$  of each  $e_i V e_j$  and let  $\mathcal{B} = \bigcup_{i,j} \mathcal{B}_{ij}$ . A non-empty word  $b_1 \cdots b_d$  in  $\mathcal{B}$  is *composable* if its adjacent idempotents match. By Lemma 2.8, these words form a basis of the positive degree part of  $T_S(V)$ .

**Definition 3.7.** *The presentation  $A = T_S(V)/(R)$  is *monomial quadratic* with respect to  $\mathcal{B}$  if  $R$  is spanned by a subset  $\mathcal{R}$  of the composable words of length 2 in  $\mathcal{B}$ .*

For  $b \in \mathcal{B}_{ij}$ , define  $b^* \in V^*$  by

$$b^*(c) = \delta_{bc} e_j \quad (c \in \mathcal{B}), \quad e_j b^* e_i = b^*. \quad (3.5)$$

The elements  $b^*$  form a basis of  $V^*$ ; their left and right idempotents are reversed.

**Proposition 3.8.** *For a monomial quadratic presentation:*

- (1) *A basis of  $A$  consists of the  $e_i$  and the non-empty composable words containing no adjacent pair from  $\mathcal{R}$ .*
- (2) *The quadratic dual is monomial, with*

$$R^\perp = \text{span}_{\mathbb{k}}\{c^* \otimes_S b^* \mid bc \text{ is composable and } bc \notin \mathcal{R}\}. \quad (3.6)$$

**Proof** The ideal generated by  $\mathcal{R}$  is spanned by the composable words containing an element of  $\mathcal{R}$  as an adjacent pair. Removing those basis words proves (1). For  $b \in \mathcal{B}_{ij}$  and  $c \in \mathcal{B}_{j\ell}$ , (3.1) and (3.5) give

$$\theta_r(c^* \otimes b^*)(b \otimes c) = c^*(e_j c) = e_\ell.$$

The functional vanishes on every other basis tensor. By Lemma 2.8, the composable  $c^* \otimes b^*$  form a basis of  $V^* \otimes_S V^*$ . Evaluating a linear combination on each  $b \otimes c \in \mathcal{R}$  forces precisely the coefficient of  $c^* \otimes b^*$  to be zero, since  $e_\ell \neq 0$ . Thus the annihilator is the span of the complementary basis tensors, proving (2).  $\square$

**Example 3.9.** *For  $A = \mathbb{k}A_n$ , take  $S = \bigoplus_i \mathbb{k}e_{ii}$  and  $V = \bigoplus_{i=1}^{n-1} \mathbb{k}e_{i,i+1}$  as in Example 2.15. There are no quadratic relations, so  $A^\dagger$  has the reversed generators  $e_{i,i+1}^* \in e_{i+1,i+1} V^* e_{ii}$  and every composable product of two generators is zero. Consequently  $h_{A^\dagger}(t) = n + (n-1)t$ . For  $n = 3$ ,*

$$(\mathbb{k}A_3)^\dagger \cong (\mathbb{k}A_3/(e_{13}))^{\text{op}}, \quad (\mathbb{k}A_3/(e_{13}))^\dagger \cong (\mathbb{k}A_3)^{\text{op}}.$$

*These are graded isomorphisms fixing the diagonal idempotents.*

**Example 3.10.** *Take  $A = T_S(V)/(ab)$  from Example 2.17, where  $S = \mathbb{k}e_1 \oplus \mathbb{k}e_2$ ,  $e_1 a e_2 = a$ , and  $e_2 b e_1 = b$ . The only composable words of length 2 are  $ab$  and  $ba$ . Thus*

$$A^\dagger = T_S(V^*)/(a^* b^*), \quad A^\dagger = S \oplus \mathbb{k}a^* \oplus \mathbb{k}b^* \oplus \mathbb{k}b^* a^*.$$

*The original relation  $ab = 0$  corresponds to the surviving word  $b^* a^*$ ; the surviving word  $ba$  corresponds to the dual relation  $a^* b^* = 0$ . In both algebras the Hilbert series is  $2 + 2t + t^2$ .*

### 3.4 Symmetric and exterior algebras

**Proposition 3.11.** *For a finite-dimensional  $\mathbb{k}$ -vector space  $V$ ,*

$$\text{Sym}(V)^\dagger \cong \bigwedge D(V), \quad (\bigwedge V)^\dagger \cong \text{Sym}(D(V)).$$

**Proof** Here  $S = \mathbb{k}$  and  $V^* = D(V)$ . Choose a basis  $x_1, \dots, x_n$  of  $V$ , with coordinate functionals  $x_1^*, \dots, x_n^*$ . The evaluation (3.1) reads

$$\theta_r(x_j^* \otimes x_i^*)(x_p \otimes x_q) = \delta_{ip} \delta_{jq}.$$

For  $F = \sum_{i,j} c_{ij} x_i^* \otimes x_j^*$ , vanishing on each  $x_p \otimes x_q - x_q \otimes x_p$  is equivalent to  $c_{pq} = c_{qp}$ . Thus the orthogonal complement of the symmetric relations is spanned by  $(x_i^*)^2$  and  $x_i^* x_j^* + x_j^* x_i^*$  for  $i < j$ , which are the exterior relations. Vanishing on the exterior relations instead gives  $c_{ii} = 0$  and  $c_{pq} + c_{qp} = 0$ . The orthogonal complement is therefore spanned by  $x_i^* x_j^* - x_j^* x_i^*$  for  $i < j$ . Now use the presentations in (2.8).  $\square$

The proof works in characteristic 2: the relations  $(x_i^*)^2 = 0$  remain part of the exterior presentation even when the signs  $+$  and  $-$  coincide. Example 2.29 is the calculation for two generators.

### 3.5 Direct sums

For graded algebras  $A, B$ , the finite direct sum  $A \oplus B$  has componentwise multiplication, unit  $(1_A, 1_B)$ , and  $(A \oplus B)_i = A_i \oplus B_i$ . In particular, its degree zero algebra is  $A_0 \oplus B_0$ .

**Proposition 3.12.** *Let  $A = T_S(V)/(R)$  and  $B = T_T(W)/(Q)$  be quadratic algebras. Then  $A \oplus B$  is quadratic over  $S \oplus T$ , and*

$$(A \oplus B)^! \cong A^! \oplus B^!.$$

**Proof** Regard  $V$  and  $W$  as  $(S \oplus T)$ -bimodules supported on the two respective factors. Mixed tensors vanish, so

$$\begin{aligned} T_{S \oplus T}(V \oplus W) &\cong T_S(V) \oplus T_T(W), \\ (V \oplus W) \otimes_{S \oplus T} (V \oplus W) &\cong (V \otimes_S V) \oplus (W \otimes_T W). \end{aligned}$$

The relations are  $R \oplus Q$ . Right module homomorphisms also split over the two factors, and their annihilator is  $R^\perp \oplus Q^\perp$ . Apply Definition 3.1.  $\square$

For example,  $(\mathbb{k}A_2 \oplus \mathbb{k}A_2)^! \cong (\mathbb{k}A_2)^{\text{op}} \oplus (\mathbb{k}A_2)^{\text{op}}$ . Both degree zero algebras are  $\mathbb{k}^{\oplus 4}$ ; there are no products connecting the two summands.

### 3.6 Tensor products and signs

**Definition 3.13.** *The **ordinary tensor product** of graded  $\mathbb{k}$ -algebras  $A, B$  has underlying vector space  $A \otimes_{\mathbb{k}} B$ , the total grading of Definition 2.24, and multiplication*

$$(a \otimes b)(a' \otimes b') = aa' \otimes bb'.$$

*The **graded tensor product**  $A \otimes_{\mathbb{k}}^{\text{gr}} B$  has the same graded vector space, with multiplication*

$$(a \otimes b)(a' \otimes b') = (-1)^{\deg b \deg a'} aa' \otimes bb' \quad \text{for homogeneous } a, a', b, b'. \quad (3.7)$$

In the ordinary tensor product,  $a \otimes 1$  and  $1 \otimes b$  commute. In the graded tensor product they satisfy

$$(1 \otimes b)(a \otimes 1) = (-1)^{\deg a \deg b} (a \otimes 1)(1 \otimes b).$$

The sign does not change the underlying graded vector space, so both tensor products have Hilbert series  $h_A(t)h_B(t)$  by Lemma 2.25.

**Proposition 3.14.** *Let  $A = T_S(V)/(R)$  and  $B = T_T(W)/(Q)$  be quadratic algebras, and assume that  $S \otimes_{\mathbb{k}} T$  is semisimple. Then both tensor products are quadratic over  $S \otimes_{\mathbb{k}} T$ , and*

$$(A \otimes_{\mathbb{k}} B)^! \cong A^! \otimes_{\mathbb{k}}^{\text{gr}} B^!, \quad (A \otimes_{\mathbb{k}}^{\text{gr}} B)^! \cong A^! \otimes_{\mathbb{k}} B^!. \quad (3.8)$$

The assumption on  $S \otimes_{\mathbb{k}} T$  holds, for example, when both  $S$  and  $T$  are products of matrix algebras over  $\mathbb{k}$ . Over a general field it does not follow from semisimplicity of  $S$  and  $T$  alone.

**Proof** The degree 1 part of  $A \otimes_{\mathbb{k}} B$  is  $(V \otimes_{\mathbb{k}} T) \oplus (S \otimes_{\mathbb{k}} W)$ . By Lemma 2.2, its tensor square over  $S \otimes_{\mathbb{k}} T$  decomposes into the four summands

$$(V \otimes_S V) \otimes_{\mathbb{k}} T \oplus \underbrace{V \otimes_{\mathbb{k}} W}_{\text{order } VW} \oplus \underbrace{V \otimes_{\mathbb{k}} W}_{\text{order } WV} \oplus S \otimes_{\mathbb{k}} (W \otimes_T W).$$

Its quadratic relations consist of  $R \otimes_{\mathbb{k}} T$ ,  $S \otimes_{\mathbb{k}} Q$ , and the mixed relations

$$(v \otimes 1)(1 \otimes w) - (1 \otimes w)(v \otimes 1) \quad (v \in V, w \in W). \quad (3.9)$$

Here the products are taken in the tensor algebra on the degree 1 part, and their span is an  $(S \otimes_{\mathbb{k}} T)$ -sub-bimodule. Let  $C$  be the quotient of that tensor algebra by the three specified relation spaces. The generator maps induce algebra homomorphisms  $A \rightarrow C$ ,  $a \mapsto \bar{a}$ , and  $B \rightarrow C$ ,  $b \mapsto \bar{b}$ . Their images commute: the degree 0 actions already commute across the two factors, and (3.9) makes the degree 1 generators commute. Consequently  $A \otimes_{\mathbb{k}} B \rightarrow C$ ,  $a \otimes b \mapsto \bar{a} \bar{b}$ , is an algebra homomorphism. It is inverse to the canonical homomorphism  $C \rightarrow A \otimes_{\mathbb{k}} B$ , since both composites fix the generating copies of  $S, T, V, W$ . Thus the stated relations give exactly  $A \otimes_{\mathbb{k}} B$ . For the graded tensor product, the same construction uses  $\bar{b} \bar{a} = (-1)^{\deg a \deg b} \bar{a} \bar{b}$  for homogeneous  $a, b$ .

The right  $(S \otimes_{\mathbb{k}} T)$ -duals of the two generator summands are  $V^* \otimes_{\mathbb{k}} T$  and  $S \otimes_{\mathbb{k}} W^*$ , with evaluations

$$(f \otimes t)(v \otimes t') = f(v) \otimes tt', \quad (s \otimes g)(s' \otimes w) = ss' \otimes g(w).$$

These identifications hold for finite free right modules and hence for their direct summands. The reversed tensor evaluation pairs each pure summand with its corresponding dual summand and exchanges the two mixed summands; all other pairings are zero. The pure relation spaces have orthogonal complements  $R^{\perp} \otimes_{\mathbb{k}} T$  and  $S \otimes_{\mathbb{k}} Q^{\perp}$  in the corresponding pure summands. The mixed relation space is the set of pairs  $(z, -z)$  in the two copies of  $V \otimes_{\mathbb{k}} W$ . For mixed tensors, the two reversed evaluations of  $f \in V^*$  and  $g \in W^*$  both give  $f(v) \otimes g(w)$  on the corresponding copy. Thus an annihilating pair of functionals must have equal components. As these simple tensors span the full dual, the mixed annihilator is spanned by

$$(f \otimes 1)(1 \otimes g) + (1 \otimes g)(f \otimes 1) \quad (f \in V^*, g \in W^*).$$

These are the mixed relations of  $A^! \otimes_{\mathbb{k}}^{\text{gr}} B^!$ , proving the first isomorphism in (3.8). For  $A \otimes_{\mathbb{k}}^{\text{gr}} B$ , the minus sign in (3.9) is a plus sign. Its annihilator has a minus sign, giving the ordinary tensor product on the dual side.  $\square$

**Example 3.15.** Take  $A = \mathbb{k}A_2 \otimes_{\mathbb{k}} \mathbb{k}A_2$ . Its degree zero algebra has basis  $e_{ii} \otimes e_{jj}$  for  $i, j \in \{1, 2\}$ , and its degree 1 part has basis

$$a = e_{12} \otimes e_{11}, \quad b = e_{22} \otimes e_{12}, \quad c = e_{11} \otimes e_{12}, \quad d = e_{12} \otimes e_{22}.$$

By Lemma 2.8, the tensor square of the degree 1 part has basis  $ab, cd$ , and both map to  $e_{12} \otimes e_{12}$ . Thus the relation space is  $\mathbb{k}(ab - cd)$ . Using the reversed generators from (3.5), its orthogonal complement is  $\mathbb{k}(b^*a^* + d^*c^*)$ . Hence the dual presentation has an anticommuting relation:

$$ab = cd \text{ in } A, \quad b^*a^* = -d^*c^* \text{ in } A^!, \quad h_A(t) = h_{A^!}(t) = 4 + 4t + t^2.$$

The signs agree with Proposition 3.14; in characteristic 2, the same presentations remain valid with  $-1 = 1$ . Here replacing  $d^*$  by  $-d^*$  removes the minus sign. In this example the sign distinguishes the specified presentations, not the isomorphism classes of algebras.

**Example 3.16.** For  $V = \mathbb{k}x \oplus \mathbb{k}y$ , Proposition 3.14 gives

$$\mathrm{Sym}(V)^! \cong (\mathbb{k}[x] \otimes_{\mathbb{k}} \mathbb{k}[y])^! \cong \frac{\mathbb{k}[x^*]}{(x^*)^2} \otimes_{\mathbb{k}}^{\mathrm{gr}} \frac{\mathbb{k}[y^*]}{(y^*)^2} \cong \bigwedge D(V).$$

If  $\mathrm{char} \mathbb{k} \neq 2$ , the ordinary tensor product of the two square-zero algebras is commutative, whereas the graded tensor product is not:  $x^*y^* = -y^*x^* \neq 0$ . They are therefore not isomorphic. This sign cannot be removed by a change of generators.

### 3.7 Dimensions and the later homological calculation

Equality of Hilbert series in Examples 3.10 and 3.15 is not a general property of quadratic duality; the symmetric and exterior series in Example 2.23 already differ. With several idempotents, it is useful to retain the dimensions of the individual blocks.

**Definition 3.17.** For a locally finite graded algebra with  $A_0 = \bigoplus_{i=1}^r \mathbb{k}e_i$ , its *matrix Hilbert series* is

$$H_A(t)_{ij} := \sum_{d \geq 0} \dim_{\mathbb{k}}(e_i A_d e_j) t^d \quad (1 \leq i, j \leq r).$$

The scalar series  $h_A(t)$  is the sum of all entries of  $H_A(t)$ . The matrix version remembers the left and right actions: (2.4) turns tensor products over  $A_0$  into matrix multiplication, while  $S$ -duality transposes the indices. For  $A = \mathbb{k}A_2$ , for example,

$$H_A(t) = \begin{pmatrix} 1 & t \\ 0 & 1 \end{pmatrix}, \quad H_{A^!}(-t)^{\mathrm{T}} = \begin{pmatrix} 1 & -t \\ 0 & 1 \end{pmatrix}, \quad H_A(t)H_{A^!}(-t)^{\mathrm{T}} = I_2.$$

In contrast,  $h_A(t)h_{A^!}(-t) = 4 - t^2$ , not 1.

The condition called *Koszul* will be defined in Lecture 5 using graded projective resolutions of  $S$ . For such algebras, Lecture 8 will prove the identity

$$H_A(t)H_{A^!}(-t)^{\mathrm{T}} = I_r \quad \text{if } A_0 = \bigoplus_{i=1}^r \mathbb{k}e_i; \quad (3.10)$$

see [BGS96, §2.11]. It is not asserted for every quadratic algebra. For the polynomial and exterior pair, the scalar instance is  $(1-t)^{-n}(1-t)^n = 1$ .

The purpose of constructing  $A^!$  is homological: for a Koszul algebra, its degree  $i$  part computes the space  $\mathrm{Ext}_A^i(S, S)$  of extensions, introduced in Lecture 6, and multiplication corresponds to joining extensions. With right modules and composition taking the right factor first, Lecture 7 will identify the resulting algebra  $\bigoplus_{i \geq 0} \mathrm{Ext}_A^i(S, S)$  with the right quadratic dual  $A^!$  over  $S$ . The choice of right  $S$ -dual is reflected already in degree 1, where  $A_1^* = \mathrm{Hom}_S(A_1, S)$  appears. No homological claim is needed for the calculations in this lecture.

### 3.8 Exercises

**Exercise 3.18.** For  $A = \mathbb{k}A_4/(e_{13})$ , write a presentation of  $A^!$  using reversed generators and find homogeneous bases of  $A$  and  $A^!$ . Check the double dual using Proposition 3.8. Hint. Among  $e_{12}e_{23}$  and  $e_{23}e_{34}$ , only the first is a relation. Complement the forbidden pairs and reverse their order.

**Exercise 3.19.** Let  $S = \mathbb{k}e_1 \oplus \mathbb{k}e_2$  and let  $V$  have basis  $a, b$  with  $e_1 a e_2 = a$  and  $e_2 b e_1 = b$ . Compute the quadratic dual of  $T_S(V)/(ab, ba)$  and the Hilbert series of both algebras. Check. The two series are  $2 + 2t$  and  $2/(1-t)$ .

**Exercise 3.20.** For the four generators in Example 3.15, replace the relation  $ab - cd$  by  $ab - \lambda cd$ , where  $\lambda \in \mathbb{k}$ . Compute the quadratic dual relation for every  $\lambda$ , including  $\lambda = 0$ , and recover the original relation by dualising again. Hint. Evaluate  $\mu b^* a^* + \nu d^* c^*$  on  $ab - \lambda cd$ . For  $\lambda \neq 0$ , rescale  $d$  to compare with Example 3.15.

## References

- [Bez23] R. Bezrukavnikov, *Noncommutative Algebra*, MIT OpenCourseWare 18.706 (2023), Lecture 2: Semisimple Modules, Socles, Artinian Rings, Wedderburn’s Theorem.
- [BGS96] A. Beilinson, V. Ginzburg, and W. Soergel, *Koszul duality patterns in representation theory*, J. Amer. Math. Soc. 9 (1996), 473–527.
- [DNN17] S. Dăscălescu, C. Năstăsescu, and L. Năstăsescu, *Graded semisimple algebras are symmetric*, J. Algebra 491 (2017), 207–218, arXiv:1504.04868.
- [GI20] J. Grant and O. Iyama, *Higher preprojective algebras, Koszul algebras, and superpotentials*, Compos. Math. 156 (2020), 2588–2627.